MA20224 Probability 2A

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Contents

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1 Events and Probabilities

In this chapter, we review some of the basic constructions from your first year probability course, but place more emphasis on some of the formal subtleties. The main goal is give meaning to the construct of a *probability space*

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

This is the basic construction that we use to model a random *experiment*. The ingredients are:

- the *sample space* Ω. Each $\omega \in \Omega$ corresponds to a realization of your experiment.
- The σ*-algebra* $\mathcal{F}$ is a collection of subsets of Ω, also known as *events*, that we would like to assign probabilities to.
- The *probability measure* $\mathbb{P}$ is a mapping that associates to each event $F \in \mathcal{F}$ a probability, i.e. a number between 0 and 1 that tells us how likely the particular event is.

In the following sections, we will look more closely at each of these ingredients. We will look at the rules they have to obey and derive some easy consequences.

1.1 Events and the Event Space

The first thing to write down, when constructing a model for a random experiment is the *sample space* Ω. Each $\omega \in \Omega$ corresponds to a particular realisation of our experiment.

Examples 1.1.

(i) Experiment C_2; toss a coin twice, so $\Omega_{C_2} = \{H, T\}^2 = \{HH, HT, TH, TT\}$.

(ii) Experiment C_∞; toss a coin infinitely many times, so $\Omega_{C_\infty} = \{H, T\}^{\mathbb{N}}$. Then $\omega \in \Omega_{C_\infty}$ is a sequence $\omega = (\omega_1, \omega_2, \dots)$ with $\omega_i \in \{H, T\}$ and ω_i is the outcome of the i-th toss for all $i \in \mathbb{N}$.

(iii) Experiment S_1; imagine we want to model the service time of the first customer in a queue, then we take $\Omega_{S_1} = \mathbb{R}^+ = (0, \infty)$.

(iv) Experiment Q; all arrival times of customers at a queue. We can take

$$\Omega_Q = \{\omega = (\omega_1, \omega_2, \dots) \,|\, \omega_i \in \mathbb{R}^+ \ \forall i, \omega_1 < \omega_2 < \cdots\} \subseteq (\mathbb{R}^+)^{\mathbb{N}},$$

where ω_i represents the arrival time of the i-th customer.

We will be interested in *events* F, that is (certain kind of) subsets of Ω.

Examples 1.2.

(i) Experiment C_2: If F is the event of getting at least one head, then $F = \{HH, HT, TH\} \subset \Omega_{C_2}$.

(ii) Experiment S_1: Let A be the event the service time takes between α and β units of time, so $A = (\alpha, \beta) \subset \Omega_{S_1}$.

(iii) Experiment C_∞; if G_p is the event the proportion of heads converges to p, then

$$G_p = \left\{ \omega = (\omega_1, \omega_2, \dots) \subset \Omega_{C_\infty} \,\middle|\, \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{\{H\}}(\omega_i) \to p \text{ as } n \to \infty \right\}$$

To formulate the last event, we have used the following notation:

Definition 1.3 (Indicator function)**.** For an event $E \subset \Omega$, the indicator function $\mathbb{1}_E$ is defined as

$$\mathbb{1}_E(\omega) := \begin{cases} 1, & \omega \in E \\ 0, & \omega \notin E \end{cases}$$

So in the above example,

$$\sum_{i=1}^{n} \mathbb{1}_{\{H\}}(w_i),$$

counts how many of the ω_i are equal to H in the first n throws of the dice. To simplify notation, we will also often write

$$\mathbb{1}_H := \mathbb{1}_{\{H\}},$$

and similarly for any other singleton sets.

Examples 1.4 (Combination of events)**.** We also frequently use that operations on events translate directly into operations on indicator functions.

Event	Description	Indicator Function
F	F occurs	$\mathbb{1}_F$
$\emptyset$	'Impossible event', nothing occurs	$\mathbb{1}_\emptyset = 0$
Ω	'Certain event', something occurs	$\mathbb{1}_\Omega = 1$
F^c	complement: F does not occur	$\mathbb{1}_{F^c} = 1 - \mathbb{1}_F$
$\bigcap_{n=1}^\infty F_n$	All F_n occur simultaneously	$\mathbb{1}_{\cap F_n} = \prod_{n=1}^\infty \mathbb{1}_{F_n}$
$\bigcup_{n=1}^\infty F_n = (\bigcap_{n=1}^\infty F_n^c)^c$	at least one F_n occurs	$1 - \prod_{n=1}^\infty (1 - \mathbb{1}_{F_n})$

To see for example the result for the intersection of events: Note that $\mathbb{1}_{\bigcap_{n=1}^\infty F_n}(\omega) = 1$ if and only if $\omega \in \bigcap_{n=1}^\infty F_n$ iff $\omega \in F_n$ for all n. In particular, for all n, we have $\mathbb{1}_{F_n}(\omega) = 1$, so that this is equivalent to $(\prod_{n=1}^\infty \mathbb{1}_{F_n})(\omega) = 1$.

In the following we will denote by $\mathcal{F}$ the collection of all events of interest for a particular experiment. If Ω is finite (or countably infinite), we usually take

$$\mathcal{F} := \mathcal{P}(\Omega) = \{A : A \subset \Omega\},$$

the *power set* of Ω. Note that if $|\Omega| = N$, then $|\mathcal{P}(\Omega)| = 2^N$. *Exercise: Why?*

Example 1.5. In the coin tossing experiment C_2, we take

$$\mathcal{F} = \mathcal{P}(\{H, T\}^2) = \{\emptyset, \{HH\}, \{HT\}, \{HH, HT\}, \dots, \Omega\},$$

Note that here $|\mathcal{F}| = 2^4 = 16$.

However, if Ω is uncountable, we cannot simply assign probabilities to all subsets of Ω. If we tried, this would lead to contradictions (look up 'non-measurable sets' or see unit 'Measure theory; also check out the Banach-Tarski paradox for some surprising consequences). The solution is not to assign probabilities to all subsets of Ω, but only to a large enough collection that contains all the events of interest. For consistency, this collection needs to satisfy the following rules:

Definition 1.6 (σ-algebra)**.** $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is a *σ-algebra* (or $\sigma-$field, or event space) on Ω if

(i) $\emptyset \in \mathcal{F}$,

(ii) if $F \in \mathcal{F}$, then $F^c \in \mathcal{F}$ and

(iii) if $(F_n : n \in \mathbb{N})$ is a sequence of sets in $\mathcal{F}$ (i.e. $F_n \in \mathcal{F}\ \forall n$), then $\bigcup_{n \in \mathbb{N}} F_n \in \mathcal{F}$.

Note that these conditions imply that $\Omega = \emptyset^c \in \mathcal{F}$ by (i) and (ii). Also, $\bigcap_n F_n = (\bigcup_n F_n^c)^c \in \mathcal{F}$ by using (ii) and (iii).

Example 1.7. As an example of very small σ-algebra consider the following. For any subset $E \subset \Omega$, we have $\mathcal{F} := \{\emptyset, E, E^c, \Omega\}$ is a σ-algebra on Ω. *Exercise: Check!*

Definition 1.8. If $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is a collection of subsets of Ω, then $\sigma(\mathcal{A})$ is known as the *σ-algebra generated by* $\mathcal{A}$ and is the smallest σ-algebra on Ω containing $\mathcal{A}$.

Note that $\sigma(\mathcal{A})$ is the intersection of all σ-algebras on Ω that contain $\mathcal{A}$. This σ-algebra is non-empty, since $\mathcal{P}(A)$ is always a σ-algebra that contains $\mathcal{A}$.

Example 1.9. Coming back to the previous example, we can check that if $E \subset \Omega$, then

$$\sigma(E) = \{\emptyset, E, E^c, \Omega\}.$$

Exercise: Can you find $\sigma(\{E, F\})$? It is already a bit more complicated!

Examples 1.10 (Important examples)**.**

(i) Let $\Omega = \mathbb{R}$. We cannot take $\mathcal{F} := \mathcal{P}(\Omega)$, since that is badly behaved. So instead we let

$$\mathcal{G} = \{(-\infty, x] : x \in \mathbb{R}\}$$

and take $\mathcal{F} = \sigma(\mathcal{G})$. This σ-algebra is known as the *Borel σ-algebra* on $\mathbb{R}$ and it denoted by $\mathcal{B}(\mathbb{R})$. It contains all subsets of $\mathbb{R}$ that are relevant for us and that we need to assign probabilities to. For example, note

(a) $(-\infty, x) = \bigcup_{n=1}^{\infty}(-\infty, x - 1/n]$, so $(-\infty, x) \in \mathcal{B}(\mathbb{R})$ by property (iii) of a σ-algebra.

(b) We could take $\mathcal{H} := \{(x, \infty) : x \in \mathbb{R}\}$ or $\mathcal{I} = \{(-\infty, x) : x \in \mathbb{R}\}$ and get the same Borel σ-algebra, i.e. $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{G}) = \sigma(\mathcal{H}) = \sigma(\mathcal{I})$.

(ii) Infinite coin tossing; for experiment C_∞, note that $|\Omega_{C_\infty}| \cong 2^{\mathbb{N}}$, which is uncountable. If $F_n := \{\omega \in \Omega_{C_\infty} \,|\, \omega_n = H\}$ is the event of getting a head on the n-th toss, then taking $\mathcal{F} = \sigma(F_n : n \in \mathbb{N})$ contains all of the events we need (including the event G_p from Example 1.2)

1.2 Probability Measures

Definition 1.11 (Kolmogorov's axioms)**.** Let Ω be a sample space and $\mathcal{F}$ a σ-algebra on Ω. Then a map $\mathbb{P} : \mathcal{F} \to [0, 1]$ is called a *probability measure* if

(i) $\mathbb{P}(\Omega) = 1$;

(ii) *σ-additivity:* If $(F_n : n \in \mathbb{N})$ is a sequence of disjoint events in $\mathcal{F}$, i.e. $F_n \in \mathcal{F}\ \forall n$ and $F_i \cap F_j = \emptyset\ \forall i \neq j$, then

$$\mathbb{P}\Big(\bigcup_{n\in\mathbb{N}} F_n\Big) = \sum_{n\in\mathbb{N}} \mathbb{P}(F_n)$$

Note a consequence of the axioms is that $\mathbb{P}(\emptyset) = 0$. We interpret $\mathbb{P}(F)$ as the probability of the event F occurring. Also, we call the triple $(\Omega, \mathcal{F}, \mathbb{P})$ a *probability space* (or a probability triple).

These axioms lead to the natural question, whether probability spaces even exist. This question is answered in the following construction of the *fundamental model.* Let $\Omega = [0, 1]$. The Borel σ-algebra on $[0, 1]$, denoted by $\mathcal{B}([0, 1])$, can be generated by $\sigma(G)$ where $G := \{[0, x] : x \in [0, 1]\}$.

Theorem 1.12 (Lebesgue, Borel)**.** *There exists a unique probability measure $\mathbb{P}$ on $\big([0, 1], \mathcal{B}([0, 1])\big)$ such that $\mathbb{P}([0, x]) = x\ \forall x \in [0, 1]$.*

The fundamental model allows us to choose a single number between $[0, 1]$ according to the uniform distribution. It is quite an amazing fact that every experiment, no matter how complicated, can be constructed out of the single experiment (as in the above theorem). [For a hint how this might work, see the problem sheets].

In the following we collect some of the properties of a probability measure. Some of these you have already seen in the first year probability course (and some will be proved on the problem sheet).

Proposition 1.13 (Properties of $\mathbb{P}$). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple, then*

(i) If $A \in \mathcal{F}$, then $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;

(ii) If $A, B \in \mathcal{F}$ and $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$;

(iii) If $A_1, \ldots, A_n \in \mathcal{F}$ are disjoint events, so $A_i \cap A_j = \emptyset \ \forall i \neq j$, then

$$\mathbb{P}\Big(\bigcup_{i=1}^n A_i\Big) = \sum_{i=1}^n \mathbb{P}(A_i)$$

(iv) Boole's inequality: *For any $A_1, \ldots, A_n \in \mathcal{F}$,*

$$\mathbb{P}\Big(\bigcup_{i=1}^n A_i\Big) \leq \sum_{i=1}^n \mathbb{P}(A_i)$$

(v) For any $A, B \in \mathcal{F}$, we have $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;

Note that the last statement also generalizes to three sets

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(B \cap C) - \mathbb{P}(A \cap C) + \mathbb{P}(A \cap B \cap C).$$

Moreover, going one step further we obtain in full generality:

Proposition 1.14. Inclusion-exclusion principle *For any $F_1, \ldots, F_n \in \mathcal{F}$,*

$$\begin{aligned}\mathbb{P}\left(\bigcup_{i=1}^n F_i\right) &= \sum_i \mathbb{P}(F_i) - \sum_{i<j} \mathbb{P}(F_i \cap F_j) + \sum_{i<j<k} \mathbb{P}(F_i \cap F_j \cap F_k) - \ldots \\ &\quad + (-1)^{n-1}\mathbb{P}(F_1 \cap \cdots \cap F_n) \\ &= \sum_{r=1}^n (-1)^{r-1} \sum_{i_1 < i_2 < \cdots < i_r} \mathbb{P}(F_{i_1} \cap \cdots \cap F_{i_r})\end{aligned}$$

Proof. We will give an elegant proof later on in Chapter 2. □

Proposition 1.15 (Continuity of $\mathbb{P}$).

(a) Continuity from below: *Consider events $A_1 \subseteq A_2 \subseteq \cdots$ and $A := \bigcup_{n \in \mathbb{N}} A_n$, then $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$. That is, the sequence $\big(\mathbb{P}(A_n) : n \in \mathbb{N}\big)$ is an increasing sequence with limit $\mathbb{P}(A)$.*

(b) Continuity from above: *Suppose* $B_1 \supseteq B_2 \supseteq \cdots$ *and* $B := \bigcap_{n\in\mathbb{N}} B_n$, *then* $\mathbb{P}(B_n) \downarrow \mathbb{P}(B)$.

Example 1.16. Consider the event $A_n := \{\text{a population of animals is extinct by time } n\}$, then $A_n \subseteq A_{n+1}$ and

$$A = \bigcup_n A_n = \{\text{population eventually becomes extinct}\}.$$

Hence, by the continuity of $\mathbb{P}$, we have that

$$\mathbb{P}(A_n) \uparrow \mathbb{P}(A).$$

2 Random Variables

When conducting a random experiment, we are often not only directly interested in the outcome, but rather in various 'observables' that can be read off from the experiment. E.g. when tossing a dice several times, we might only be interested in the sum of all the throws. That is we want to understand mappings from Ω to $\mathbb{R}$, since we are typically making a real-valued observation. This naturally leads to the concept of a random variable. Indeed, working on the same probability space we might also be interested in understanding different observables (e.g. also the difference of the first two throws of the dice). Later on in the course, we might often 'forget' about the underlying probability space and write our assumptions in terms of random variables only.

Definition 2.1. A random variable X on $\mathbb{R}$ is a mapping $X : \Omega \to \mathbb{R}$ such that

$$\{X \leq x\} := \{\omega \in \Omega \,:\, X(\omega) \leq x\} \in \mathcal{F} \;\; \forall x \in \mathbb{R} \tag{1}$$

Remark 2.2. (i) Note that since $\mathcal{B}(\mathbb{R}) = \sigma\big(\{(-\infty, x] : x \in \mathbb{R}\}\big)$ and $\mathcal{F}$ is a σ-algebra, then one can show that

$$X^{-1}(A) := \{X \in A\} := \{\omega \in \Omega \,|\, X(\omega) \in A\} \in \mathcal{F} \;\text{ for any } A \in \mathcal{B}(\mathbb{R}) \tag{2}$$

We say that X is an $\mathcal{F}$-measurable function from Ω to $\mathbb{R}$.

(ii) Often we want random variables that can take values on $\mathbb{R} \cup \{+\infty, -\infty\}$ or $\mathbb{R}^+ \cup \{+\infty\}$. The definition above extends simply by replacing $\mathbb{R}$ with these.

(iii) Note: everything that you are likely to meet in probability will be suitably measurable! So you should be aware of these problems, but you don't have to worry too much about these.

Example 2.3. Consider Experiment Q from Example 1.1, where we model the arrival times of customers in a queue. So if $\omega = (\omega_1, \omega_2, \dots) \in \Omega_Q$, and we are interested in T_1, the time of the first arrival at the queue, then we can formally define $T_1 : \Omega \to \mathbb{R}$ as $T_1(\omega) = \omega_1$. Another example might be the gap between the arrival of the first and second customers: $G_1 : \Omega \to \mathbb{R}$, is given by $G_1(\omega) := \omega_2 - \omega_1$.

In order to describe a random variable, we introduce the concept of its distribution.

Definition 2.4. The *distribution* of a random variable X is the probability measure $\mathbb{P}_X$ defined on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ via

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A) \qquad \text{for any } A \in \mathcal{B}(\mathbb{R}).$$

The *distribution function* F_X of a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$ is the map $F_X : \mathbb{R} \to [0, 1]$ defined by

$$F(x) := \mathbb{P}(X \leq x) = \mathbb{P}\big(\{\omega \in \Omega : X(\omega) \leq x\}\big)$$

Notice that in order for these quantities to make sense, we are really using that X is a random variable so that (2) and (1) hold. As before in Remark 2.2 it turns out that the distribution function completely describes the distribution of a random variable.

On the problem sheet, we will see that any distribution function F is an increasing, right-continuous function such that $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$. Conversely, one can show that any function F with these properties is the distribution function of a real-valued random variable.

Examples 2.5.

(i) We say that T_1 is exponentially distributed with rate $q > 0$, or $T_1 \sim \mathrm{Exp}(q)$, if

$$F_{T_1}(t) = \mathbb{P}(T_1 \le t) = \begin{cases} 1 - e^{-qt}, & t \ge 0 \\ 0, & t < 0 \end{cases}$$

(ii) Suppose X is a random variable taking values $\{1, 2, 3\}$ with equal probability, then

$$F_X(x) = \mathbb{P}(X \le x) = \begin{cases} 0, & x < 1 \\ 1/3, & 1 \le x < 2 \\ 2/3, & 2 \le x < 3 \\ 1, & x \ge 3 \end{cases}$$

There are two cases when its particular easy to describe the distribution of a random variable.

Definition 2.6. Suppose the random variable X takes values in a countable set, $X(\Omega) := \{X(\omega) : \omega \in \Omega\}$. Then we say that X is a *discrete* random variable and we call the function

$$p_X(x) := \mathbb{P}(X = x) = \mathbb{P}\big(\{\omega \in \Omega : X(\omega) = x\}\big),$$

where $p_X : X(\Omega) \to [0, 1]$, the *probability mass function* (PMF) of X.

Example 2.7. In the Example 2.5(ii)

$$p_X(x) = \begin{cases} 1/3, & x \in \{1, 2, 3\} = X(\Omega) \\ 0, & \text{otherwise} \end{cases}$$

Definition 2.8. A random variable is called *continuous* if there exists a non-negative function f_X on $\mathbb{R}$ such that

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a) = \int_a^b f_X(x)\,\mathrm{d}x$$

where f_X is called the *probability density function* (PDF) of X.

Example 2.9. (i) If $T_1 \sim \mathrm{Exp}(q)$ as in Example 2.5(i), then one can check that it is continuous with density

$$f_{T_1}(t) = \begin{cases} qe^{-qt}, & t \ge 0 \\ 0, & t < 0 \end{cases}.$$

(ii) The following is also a distribution function of a random variable X,

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ \frac{1}{2}x & \text{if } x \in (0,1) \\ 1 & \text{if } x \geq 1 \end{cases}$$

The correspoding random variable is neither discrete nor continuous (see also the problem sheet).

2.1 Expectation

Definition 2.10. Let X be a random variable with $0 \leq X \leq \infty$. If X is discrete, then

$$\mathbb{E}(X) := \sum_{x \in X(\Omega)} x\, \mathbb{P}(X = x) = \sum_{x \in X(\Omega)} x\, p_X(x) \leq \infty$$

Note that an indicator function $\mathbb{1}_E$ of an event E is a discrete random variable (taking values 0 and 1) and we have that

$$\mathbb{E}(\mathbb{1}_E) = 0 \cdot \mathbb{P}(\mathbb{1}_E = 0) + 1 \cdot \mathbb{P}(\mathbb{1}_E = 1) = \mathbb{P}(E)$$

since $\mathbb{1}_E(\omega) = 1$ if, and only if, $\omega \in E$.

Definition 2.11. For any random variable X with $0 \leq X \leq \infty$, we define

$$\mathbb{E}(X) := \int X(\omega)\, \mathrm{d}\mathbb{P}(\omega) := \sup\{\mathbb{E}(Y) : Y \text{ is discrete and } Y \leq X\} \in [0, \infty].$$

As you will see later in the unit 'Measure theory' the expectation is thus the Lebesgue integral of X with respect to $\mathbb{P}$.

As a consequence of the definition, one can show that

$$\mathbb{E}(X) = \begin{cases} \displaystyle\sum_{x \in X(\Omega)} x\, \mathbb{P}(X = x), & \text{if } X \text{ discrete} \\ \displaystyle\int_{\mathbb{R}} x\, f_X(x)\, \mathrm{d}x, & \text{if } X \text{ continuous} \end{cases}$$

Lemma 2.12 (Properties of expectation)**.**

(i) If $X, Y \geq 0$ and $\lambda, \mu > 0$ are constants, then $\mathbb{E}(\lambda X + \mu Y) = \lambda \mathbb{E}(X) + \mu \mathbb{E}(Y)$.

(ii) If $X \geq 0$ and $\mathbb{P}(X = \infty) > 0$, then $\mathbb{E}(X) = \infty$; so

(ii)' If $X \geq 0$ and $\mathbb{E}(X) < \infty$, then $\mathbb{P}(X < \infty) = 1$ (contrapositive).

Note that if an event holds with probability 1, then we also say that it holds *almost surely.* I.e. another way of saying $\mathbb{P}(X < \infty) = 1$ is 'X is finite almost surely'.

Theorem 2.13 (Monotone convergence theorem, MON). *If $(X_n)_{n\in\mathbb{N}}$ is a sequence of non-negative random variables, then*

$$0 \le X_n \uparrow X \Longrightarrow \mathbb{E}(X_n) \uparrow \mathbb{E}(X) \ \ (\le \infty)$$

That is, if for every $\omega \in \Omega$, $X_n(\omega) \le X_{n+1}(\omega)$ $\forall n \in \mathbb{N}$ and $X(\omega) = \lim_{n\to\infty} X_n(\omega)$, then $\lim_{n\to\infty} \mathbb{E}(X_n) = \mathbb{E}\left(\lim_{n\to\infty} X_n\right)$, i.e. $\mathbb{E}(X_n) \uparrow \mathbb{E}(X)$.

Remark 2.14. If $F_1 \subseteq F_2 \subseteq \cdots$ and $F := \bigcup_n F_n$, then $\mathbb{1}_{F_n} \uparrow \mathbb{1}_F$ so $\mathbb{E}(\mathbb{1}_{F_n}) \uparrow \mathbb{E}(\mathbb{1}_F)$ by the monotone convergence theorem, that is, $\mathbb{P}(F_n) \uparrow \mathbb{P}(F)$. Hence MON implies the continuity of probability measures.

Proposition 2.15. *Suppose that $X_k \ge 0$, then*

$$\sum_{k=1}^{\infty} \mathbb{E}(X_k) < \infty \implies \mathbb{E}\Big(\sum_{k=1}^{\infty} X_k\Big) = \sum_{k=1}^{\infty} \mathbb{E}(X_k) < \infty$$

$$\implies \mathbb{P}\Big(\sum_{k=1}^{\infty} X_k < \infty\Big) = 1 \implies \mathbb{P}(X_k \to 0) = 1$$

Proof. For the first implication, note that

$$X_k \ge 0 \Longrightarrow \sum_{k=1}^{n} X_k \uparrow \sum_{k=1}^{\infty} X_k$$

so MON gives

$$\mathbb{E}\Big(\sum_{k=1}^{n} X_k\Big) \uparrow \mathbb{E}\left(\sum_{k=1}^{\infty} X_k\right)$$

but on the other hand by the linearity of expectation, Lemma 2.12,

$$\mathbb{E}\Big(\sum_{k=1}^{n} X_k\Big) = \sum_{k=1}^{n} \mathbb{E}(X_k) \uparrow \sum_{k=1}^{\infty} \mathbb{E}(X_k)$$

by definition of the infinite sum, since $\mathbb{E}(X_k) \ge 0$. Hence, we have shown that

$$\mathbb{E}\Big(\sum_{k=1}^{\infty} X_k\Big) = \sum_{k=1}^{\infty} \mathbb{E}(X_k).$$

The second implication follows from Lemma 2.12(ii)'

For the third implication, note that for any ω, $X_k \ge 0$ and $\sum_{k\ge1} X_k < \infty$ implies that $X_k \to 0$ as $k \to \infty$. □

This proposition has a very important corollary.

Lemma 2.16 (First Borel-Cantelli lemma)**.** *Let $J_1, J_2, \dots$ be a sequence of events. If $\sum_{k=1}^{\infty} \mathbb{P}(J_k) < \infty$, then it is almost surely true that only finitely many of the events J_k occur.*

Proof. Let $X_k = \mathbb{1}_{J_k}$, then $Y = \sum_{k=1}^{\infty} X_k$ counts the total number of successes in all events. Since $\sum_{k=1}^{\infty} \mathbb{E}(X_k) = \sum_{k=1}^{\infty} \mathbb{P}(J_k) < \infty$, we have that $\mathbb{E}(Y) < \infty$ (first implication of Proposition 2.15) and hence $Y = \sum_{k=1}^{\infty} \mathbb{1}_{J_k} < \infty$ by Lemma 2.12(ii)'. That is, we have a finite total number of successes with probability one. $\square$

2.2 Integrable X, $\mathcal{L}^1$

So far we have only defined the expected value for non-negative random variables. To extend it to general random variables, we use the following construction. Suppose $X : \Omega \to \mathbb{R}$ is a random variable on $\mathbb{R}$. Define the positive, resp. negative, part of X as

$$X^+(\omega) := \max\{0, X(\omega)\} \quad \text{ and } X^-(\omega) := \max\{0, -X(\omega)\}.$$

Then X^+ and X^- are both non-negative random variables and we have that

$$X = X^+ - X^- \quad \text{ and } \quad |X| = X^+ + X^-$$

(where the equality holds for all $\omega \in \Omega$.)

Definition 2.17. If both $\mathbb{E}(X^+)$ and $\mathbb{E}(X^-)$ are finite, equivalently, if $\mathbb{E}|X| = \mathbb{E}(X^+) + \mathbb{E}(X^-) < \infty$, we call X *integrable*, denoted as $X \in \mathcal{L}^1$. In this case, we define

$$\mathbb{E}(X) = \mathbb{E}(X^+) - \mathbb{E}(X^-).$$

Proposition 2.18 (Properties of the expectation)**.**

(i) If $X \in \mathcal{L}^1$, then $|\mathbb{E}(X)| \leq \mathbb{E}(|X|)$;

(ii) If $X, Y \in \mathcal{L}^1$ and $\lambda, \mu \in \mathbb{R}$, then $\lambda X + \mu Y \in \mathcal{L}^1$ and $\mathbb{E}(\lambda X + \mu Y) = \lambda \mathbb{E}(X) + \mu \mathbb{E}(Y)$;

(iii) If X is a discrete random variable and $h : X(\Omega) \to \mathbb{R}$, then

$$\mathbb{E}\big(h(X)\big) = \sum_{x \in X(\Omega)} h(x)\, \mathbb{P}(X = x)$$

where $h(X)$ is considered integrable if, and only if, the series converges absolutely.

(iv) If X is continuous with PDF f_X, then for any piecewise continuous function $h : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}\big(h(X)\big) = \int_{\mathbb{R}} h(x)\, \mathbb{P}(X \in \mathrm{d}x) := \int_{\mathbb{R}} h(x) f_X(x)\, \mathrm{d}x$$

where $h(X)$ is integrable if, and only if,

$$\int_{\mathbb{R}} |h(x)| f_X(x)\, \mathrm{d}x < \infty$$

Proof. See the unit 'Measure theory' or any textbook on probability. $\square$

Example 2.19. Suppose $T_1 \sim \text{Exp}(q)$, then we have that

$$\mathbb{E}(T_1) = \int_{-\infty}^{\infty} t f_{T_1}(t)\, dt = \int_0^{\infty} tqe^{-qt}\, dt = \frac{1}{q},$$

where in order to calculate the last integral we use integration by parts (Details: *Exercise!*).

We will now give an easy proof of the inclusion-exclusion principle using expectations and the fact that the expectation of an indicator function $\mathbb{1}_E$ is just the probability $\mathbb{P}(E)$. First, recall the statement of the inclusion-exclusion principle, Proposition 1.14: For any $F_1, \dots, F_n \in \mathcal{F}$,

$$\begin{aligned}\mathbb{P}\left(\bigcup_{i=1}^n F_i\right) &= \sum_i \mathbb{P}(F_i) - \sum_{i<j} \mathbb{P}(F_i \cap F_j) + \sum_{i<j<k} \mathbb{P}(F_i \cap F_j \cap F_k) - \dots \\ &\quad + (-1)^{n-1}\mathbb{P}(F_1 \cap \dots \cap F_n) \\ &= \sum_{r=1}^n (-1)^{r-1} \sum_{i_1<i_2<\dots<i_r} \mathbb{P}(F_{i_1} \cap \dots \cap F_{i_r})\end{aligned}$$

Proof. We have that

$$\mathbb{1}_{\bigcup_{i=1}^n F_i} = 1 - \mathbb{1}_{\bigcap_{i=1}^n F_i^c} = 1 - \prod_{i=1}^n \mathbb{1}_{F_i^c} = 1 - \prod_{i=1}^n (1 - \mathbb{1}_{F_i})$$

and for $x_i \in [0,1]$,

$$\prod_{i=1}^n (1 - x_i) = 1 - \sum_i x_i + \sum_{i<j} x_i x_j - \sum_{i<j<k} x_i x_j x_k + \dots + (-1)^n x_1 x_2 \cdots x_n,$$

hence

$$\begin{aligned}\mathbb{P}\left(\bigcup_{i=1}^n F_i\right) &= \mathbb{E}\left(\mathbb{1}_{\bigcup_{i=1}^n F_i}\right) = \mathbb{E}\left[\sum_i \mathbb{1}_{F_i} - \sum_{i<j} \mathbb{1}_{F_i}\mathbb{1}_{F_j} + \dots - (-1)^n\, \mathbb{1}_{F_1} \cdots \mathbb{1}_{F_n}\right] \\ &= \sum_i \mathbb{E}\left(\mathbb{1}_{F_i}\right) - \sum_{i<j} \mathbb{E}\left(\mathbb{1}_{F_i \cap F_j}\right) + \dots + (-1)^{n-1}\mathbb{E}\left(\mathbb{1}_{F_1 \cap \dots \cap F_n}\right) \\ &= \sum_i \mathbb{P}(F_i) - \sum_{i<j} \mathbb{P}(F_i \cap F_j) + \sum_{i<j<k} \mathbb{P}(F_i \cap F_j \cap F_k) - \dots \\ &\quad + (-1)^{n-1}\mathbb{P}(F_1 \cap \dots \cap F_n),\end{aligned}$$

as claimed. $\square$

Theorem 2.20 (Bounded-convergence theorem)**.** *Suppose* $(X_n)_{n\in\mathbb{N}}$ *and* X *are random variables such that* $X_n \to X$ *almost surely, that is,* $\mathbb{P}(X_n \to X \text{ as } n \to \infty) = 1$*. Suppose further that* $\exists K > 0$ *such that*

$$|X_n(\omega)| \leq K, \quad \forall n, \forall \omega \in \Omega.$$

Then

$$\mathbb{E}(|X_n - X|) \to 0$$

and hence

$$\mathbb{E}(X_n) \to \mathbb{E}(X).$$

Proof. Assumed; for the last part we note that $-|X_n - X| \leq X_n - X \leq |X_n - X|$. $\square$

3 Conditional probabilities and independence

In this chapter, we recall the basic concepts of conditional probabilities and independence and derive the complement to the first Borel-Cantelli lemma. Throughout, we let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

3.1 Conditional Probabilities

Definition 3.1. Suppose that $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$, then $\forall B \in \mathcal{F}$, define

$$\mathbb{P}(B|A) := \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)}$$

We can check that in this case, $\mathbb{P}(\cdot|A) : \mathcal{F} \to [0, 1]$ is a probability measure on $(\Omega, \mathcal{F})$.

You have already seen the next two results in the first year probability course. First recall that from the definition if $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\, \mathbb{P}(B \mid A). \tag{3}$$

If $\mathbb{P}(A \cap B) > 0$, this generalizes to three events:

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\, \mathbb{P}(B \mid A)\, \mathbb{P}(C \mid A \cap B).$$

To see that this holds, write $A \cap B \cap C = (A \cap B) \cap C$ and apply (3) twice. Even more generally, we have:

Theorem 3.2 (Multiplication principle). *Suppose $A_i \in \mathcal{F}$ for all $i \in \{1, \ldots, n\}$. If $\mathbb{P}(A_1 \cap \cdots \cap A_n) > 0$, then $\mathbb{P}(A_1 \cap \cdots \cap A_j) > 0\ \forall j = 1, \ldots, n-1$ and*

$$\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\mathbb{P}(A_2|A_1)\mathbb{P}(A_3|A_1 \cap A_2) \cdots \mathbb{P}(A_n|A_1 \cap \cdots \cap A_{n-1})$$

The multiplication principle leads to the well-known *Bayes' theorem* as follows.

Suppose $H_1, \ldots, H_n \in \mathcal{F}$ are a partition of Ω in that they are disjoint

$$H_i \cap H_j = \emptyset \text{ for all} i \neq j,$$

and cover Ω in that

$$\bigcup_{k=1}^{n} H_k = \Omega.$$

Assume that $\mathbb{P}(H_k) > 0\ \forall k$. Let $K \in \mathcal{F}$, then from the additivity of probability measures and the multiplication principle (3), we have the *law of total probability:*

$$\mathbb{P}(K) = \sum_{j=1}^{n} \mathbb{P}(H_j \cap K) = \sum_{j=1}^{n} \mathbb{P}(H_j)\mathbb{P}(K|H_j) \tag{4}$$

Now, if additionally $\mathbb{P}(K) > 0$, then again by (3)

$$\mathbb{P}(H_i|K) = \frac{\mathbb{P}(H_i \cap K)}{\mathbb{P}(K)} = \frac{\mathbb{P}(H_i)\mathbb{P}(K|H_i)}{\mathbb{P}(K)}.$$

Substituting in (4), we have proved:

Theorem 3.3 (Bayes' theorem)**.** *Let $H_1, \ldots, H_n \in \mathcal{F}$ be a partition of Ω with $\mathbb{P}(H_i) > 0$. Then, for any $K \in \mathcal{F}$ with $\mathbb{P}(K) > 0$, we have*

$$\mathbb{P}(H_i|K) = \frac{\mathbb{P}(H_i)\mathbb{P}(K|H_i)}{\sum_{j=1}^{n} \mathbb{P}(H_j)\mathbb{P}(K|H_j)}.$$

3.2 Independence

A central concept in probability is independence. We recall the definition for events and then see how it applies to random variables.

Recall, events $A, B \in \mathcal{F}$ are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

If A and B are independent, then $\mathbb{P}(A|B) = \mathbb{P}(A|B^c) = \mathbb{P}(A)$, so whether A occurs or not does not affect the probability of A occurring. The concept generalizes to finitely many and also infinitely many events.

Definition 3.4. (i) *Finitely many events.* Events $E_1, E_2, \ldots, E_n$ are *independent* if for any subsequence $i_1 < i_2 < \ldots < i_k$ of $\{1, \ldots, n\}$ we have

$$\mathbb{P}(E_{i_1} \cap \ldots \cap E_{i_k}) = \mathbb{P}(E_{i_1}) \cdots \mathbb{P}(E_{i_k}).$$

(ii) *Infinitely many events.* Events $E_1, E_2, \ldots$ are *independent* if for every $n \in \mathbb{N}$, the events $E_1, \ldots, E_n$ are independent.

With the extra assumption of independence we can formulate the complementary result to the first Borel-Cantelli lemma.

Lemma 3.5 (Second Borel-Cantelli lemma)**.** *Suppose $J_1, J_2, \ldots$ is an infinite sequence of independent events such that $\sum_{i=1}^{\infty} \mathbb{P}(J_i) = +\infty$. Then the probability that infinitely many J_n occur is 1, that is,*

$$\mathbb{P}\Big(\sum_{n=1}^{\infty} \mathbb{1}_{J_n} = +\infty\Big) = 1$$

Proof. We have that

$$\begin{aligned}\mathbb{P}(\text{at least one of } J_1, \ldots, J_n \text{ occur}) &= \mathbb{P}\left(\bigcup_{i=1}^{n} J_i\right) = 1 - \mathbb{P}\left(\bigcap_{i=1}^{n} J_i^c\right) \\ &= 1 - \mathbb{P}(\text{none of } J_1, \ldots, J_n \text{ occur})\end{aligned}$$

and

$$\mathbb{P}\left(\bigcap_{i=1}^{n} J_i^c\right) \stackrel{*}{=} \prod_{i=1}^{n} \mathbb{P}(J_i^c) = \prod_{i=1}^{n} \big(1 - \mathbb{P}(J_i)\big) \leq \exp\left(-\sum_{i=1}^{n} \mathbb{P}(J_i)\right)$$

where $(*)$ follows by independence and $1 - x \leq e^{-x}\ \forall x$. Now we have that

$$\bigcap_{i=1}^{n} J_i^c \downarrow \bigcap_{i=1}^{\infty} J_i^c$$

so by MON/continuity of probability, this implies that

$$\mathbb{P}\left(\bigcap_{i=1}^{n} J_i^c\right) \downarrow \mathbb{P}\left(\bigcap_{i=1}^{\infty} J_i^c\right) = \mathbb{P}(\text{none of } J_1, J_2, \dots \text{ occur})$$

but

$$0 \leq \mathbb{P}\left(\bigcap_{i=1}^{n} J_i^c\right) \leq \exp\left(-\sum_{i=1}^{n} \mathbb{P}(J_i)\right) \to 0 \text{ as } n \to \infty$$

since $\sum_{i=1}^{\infty} \mathbb{P}(J_i) = +\infty$ by assumption. So

$$\mathbb{P}(\text{none of } J_1, J_2, \dots \text{ occur}) = 0 \Longrightarrow \mathbb{P}(\text{at least one of } J_1, J_2, \dots \text{ occur}) = 1$$

Proof will be continued on Problem Sheet 2. □

Definition 3.6. A finite or infinite sequence of random variables is *independent* if for any $x_1, x_2, x_3, \dots \in \mathbb{R}$, the events

$$\{X_1 \leq x_1\}, \{X_2 \leq x_2\}, \{X_3 \leq x_3\}, \dots$$

are independent.

Remark 3.7. By the definition of the independence of events this means that whenever $i_1, i_2, \dots, i_r$ are distinct integers such that X_{i_k} exist (in the case of finite random variables), and $x_1, \dots, x_r \in \mathbb{R}$ we have

$$\mathbb{P}(X_{i_1} \leq x_1; X_{i_2} \leq x_2; \dots; X_{i_r} \leq x_r) = \mathbb{P}(X_{i_1} \leq x_1)\mathbb{P}(X_{i_2} \leq x_2) \cdots \mathbb{P}(X_{i_r} \leq x_r)$$

We will often use, but not prove, the following fact.

Proposition 3.8. *If $X_1, X_2, \dots$ are independent random variables and $f_1, f_2, \dots$ are nice[2] functions on $\mathbb{R}$, then $f_1(X_1), f_2(X_2), \dots$ are independent. Moreover, if $g : \mathbb{R}^n \to \mathbb{R}$ is a nice function, then the random variables $g(X_1, \dots, X_n), X_{n+1}, X_{n+2}, \dots$ are independent.*

[2] The technical term is Borel measurable, see also Remark 3.10. Most functions you will come across are Borel measurable. Continuous functions would work for example.

An important property of independent random variables is that they make working out the expectation of a product particularly easy. The more advanced version of a result you have seen in year 1 probability:

Lemma 3.9. *Let X, Y be independent random variables, then*

(i) If $X \geq 0, Y \geq 0$, then $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$;

(ii) If $X, Y \in \mathcal{L}^1$, then $XY \in \mathcal{L}^1$ and $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$.

Remark 3.10. Technical clarification [Not examinable!]: Let S be a topological space (e.g. $\mathbb{R}$ or $\mathbb{R}^n$). Then the Borel σ-algebra $\mathcal{B}(S)$ is the smallest σ-algebra that contains all the open subsets of S. A *Borel subset* of S is an element of $\mathcal{B}(S)$.

A function $f : \mathbb{R}^n \to \mathbb{R}$ is *Borel measurable* if for any $x \in \mathbb{R}$,

$$\{s \in \mathbb{R}^n \,:\, f(s) \leq x\} \in \mathcal{B}(\mathbb{R}^n).$$

For the more general definition, we need some basics about topological spaces (which we are not assuming in this unit): Recall that a *continuous* function $f : S \to \mathbb{R}$ is one such that the inverse image

$$f^{-1}(B) = \{s \in S \,:\, f(s) \in B\},$$

of any open subset $B \subset \mathbb{R}$ is an open subset of S.

Analogously, $f : S \to \mathbb{R}$ is *Borel measurable* if the inverse image $f^{-1}(B)$ of any Borel subset $B \subset \mathbb{R}$ is Borel in S. Equivalently, if for any $x \in \mathbb{R}$,

$$\{s \in S \,:\, f(s) \leq x\} \in \mathcal{B}(S).$$

4 Random Walks

In this chapter, we will introduce the first basic probabilistic model, a *random walk*, which we will be studying throughout. In the easiest case, this is a process that at each step either takes one step upwards or one step downwards. In this chapter, we will first determine the probability that the random walker hits 0 when started from any other state. Secondly, we will consider the gambler's ruin problem, where we work out the probability that the random walk hits a state N before hitting 0. We will also find the expected value of the first time that either state 0 or N are hit.

4.1 Simple Random Walks: Hitting and return probabilities

The simplest case is a random walk on $\mathbb{Z}$, where in each step a 'particle' independently decides to go up (with probability p) and down with probability $1-p$.

More formally, let $p \in (0,1)$, then the position of a *simple random walk* SRW(p) at time n is given by

$$W_n := a + X_1 + X_2 + \cdots + X_n$$

where $W_0 := a \in \mathbb{Z}$ is the starting position and $(X_n : n \in \mathbb{N})$ is an independent and identically distributed (i.i.d.) sequence of random variables with

$$\mathbb{P}(X_n = +1) = p, \;\; \mathbb{P}(X_n = -1) = q := 1-p$$

This is illustrated in Figure 1.

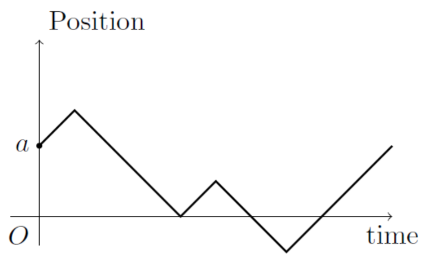


Figure 1: A random walk trajectory.

Note that probabilities for this random walk $W = (W_n)_{n \geq 0}$ will depend on the starting position a: we write $\mathbb{P}_a$ for the probability law for the random walk when $W_0 = a$ and $\mathbb{E}_a$ for the corresponding expectation.

In the following we will derive the *hitting/return probabilities* of 0 for a simple random walk. Let H be the event that $W_n = 0$ for some strictly positive time,

$$H := \{W_n = 0 \text{ for some } n \geq 1\}$$

and define $x_i := \mathbb{P}_i(H)$.

If $i \neq 0$, then x_i is the probability the random walk started at position i and hits 0; but if $i = 0$, then x_0 is the probability the random walk started at position 0 and returns to the origin.

Now we have by the law of total probability that if we split according to the first step

$$\begin{aligned}\mathbb{P}_i(H) &= \mathbb{P}(H|W_0 = i) = \mathbb{P}_i(H; X_1 = +1) + \mathbb{P}_i(H; X_1 = -1)\\ &= \mathbb{P}_i(X_1 = +1)\mathbb{P}_i(H|X_1 = +1) + \mathbb{P}_i(X_1 = -1)\mathbb{P}_i(H|X_1 = -1)\end{aligned}$$

We start by considering the case when $W_0 = +1$:

$$\mathbb{P}_1(H) = p\,\mathbb{P}_1(H|X_1 = +1) + q\,\mathbb{P}_1(H|X_1 = -1)$$

but $\mathbb{P}_1(H|X_1 = +1) = \mathbb{P}_2(H)$ as conditioning on $X_1 = +1$ takes the random walk to position 2 and then everything 'starts afresh' at time 1 since $X_2, X_3, \ldots$ are independent of X_1 and $\mathbb{P}_1(H|X_1 = -1) = 1$ since conditional on $X_1 = -1$, we have $W_1 = 0$ and event H is guaranteed to occur. Thus

$$\begin{cases} x_1 = px_2 + q, & \\ x_i = px_{i+1} + qx_{i-1} & \text{for all } i \geq 2 \end{cases} \tag{5}$$

Note that $x_1 \neq px_2 + qx_0$ since x_0 need not be 1, whereas the random walk jumping down from 1 to 0 on the first step definitely hits 0 at a positive time.

Intuitively,

$$\begin{aligned}x_2 &= \mathbb{P}_2(\text{hit } 0)\\ &= \mathbb{P}_2(\text{hit } 1)\mathbb{P}_2(\text{hit } 0 \,|\, \text{hit } 1)\\ &= \mathbb{P}_2(\text{hit } 1)\mathbb{P}_1(\text{hit } 0),\end{aligned}$$

since once the random walk gets to 1 from 2, everything 'starts afresh' and independently it must get down from 1 to 0. Formally, we are using the *strong Markov property* of the random walk at random time $T_1 = \inf\{n \geq 0 : W_n = 1\}$, but we will not discuss this in all formal details here, see also Figure 2.

Thus, since the probability of going from 2 to 1 must be the same as from 1 to 0, we have shown

$$x_2 = x_1^2. \tag{6}$$

Similarly, we find that

$$x_i = (x_1)^i \qquad \text{for } i \geq 1. \tag{7}$$

Combining (5) and (6) we have that

$$x_1 = px_1^2 + q \Longrightarrow px_1^2 - x_1 + q = (px_1 - q)(x_1 - 1) = 0.$$

Hence, $x_1 = q/p$ or $x_1 = 1$.

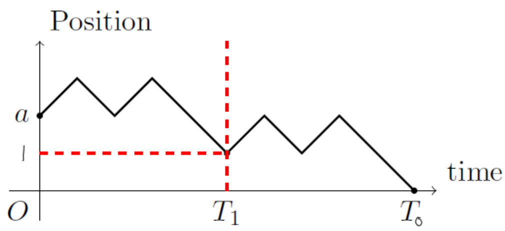


Figure 2: The random walk starts afresh once reaching level 1.

If $q \geq p$, only one of the solutions is in $[0, 1]$, so we must have $x_1 = 1$. Thus, we are guaranteed to return if we are more likely to jump down.

If $q < p$, both solutions could be probabilities. Intuitively, since $\mathbb{E}(X_k) = 1.p + (-1).q = p - q$ we will have

$$\frac{W_n}{n} = \frac{W_0 + X_1 + X_2 + \cdots + X_n}{n} \to \mathbb{E}(X_1) = p - q > 0$$

(this is the strong law of large numbers, SLLN, that we will discuss later) and hence W_n tends to $+\infty$, so there must be a positive probability of drifting off to $+\infty$ without hitting 0. Hence if $q < p$, then $x_1 = q/p$.

Combining this discussion with (7), we have shown the following:

Theorem 4.1. *Let* (W_n) *be a simple random walk with* $q \leq p$. *Then,*

$$\mathbb{P}_i(hit\ 0) = \mathbb{P}_i(H) = \begin{cases} (q/p)^i, & i \geq 1 \\ 1, & i \leq -1 \end{cases}$$

Note that for $p < q$ an analogous result holds by symmetry.

Lastly, noting

$$x_0 = px_1 + qx_{-1} = \begin{cases} p(q/p) + q.1 = 2q, & q \leq p \\ p.1 + q(p/q) = 2p, & p \leq q \end{cases}$$

we find that $x_0 = 1 - |p - q|$ for any $p \in (0, 1)$, hence in all cases

$$\mathbb{P}_0(\text{no return to } 0) = 1 - x_0 = |p - q|$$

which is the bias in the random walk.

4.2 Gambler's Ruin

For the *gambler's ruin problem*, we consider a gambler who consecutively plays the following game: at each step she bets $\mathbb{P}ounds1$: if she wins then she receives $\mathbb{P}ounds2$ in return (i.e. she gains $\mathbb{P}ounds1$) and otherwise she loses her pound. If she starts with k pounds and wins a game with probability p, then the evolution of her capital is described by a simple random walk. In this section, we answer the question, how likely it is that she gains $\mathbb{P}oundsN$ before going bankrupt.

Let k and N be integers with $0 < k < N$. Let

$$x_k := \mathbb{P}_k(T_N < T_0),$$

where

$$T_j := \inf\{n \geq 0 : W_n = j\},$$

is the first hitting time of state j for the SRW(p) with $W = (W_n)_{n\geq 1}$. That is, x_k is the probability the SRW(p) that started at k will hit N before 0.

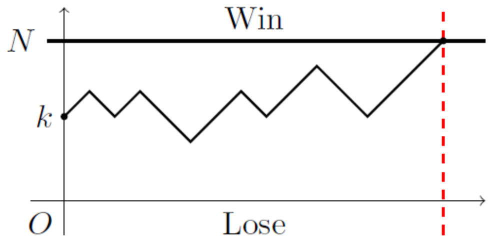


Figure 3: Gambler's ruin.

Considering the first move as before, we have that for $1 \leq k \leq N-1$,

$$x_k = px_{k+1} + qx_{k-1} \qquad \text{where } x_0 = 0,\, x_N = 1$$

which is the recurrence/difference equation. Since $p+q = 1$, we can rewrite this equation as

$$p(x_{k+1} - x_k) - q(x_k - x_{k-1}) = 0.$$

and therefore

$$x_{k+1} - x_k = \frac{q}{p}(x_k - x_{k-1}) = \left(\frac{q}{p}\right)^2 (x_{k-1} - x_{k-2})$$

Continuing, we obtain

$$x_{k+1} - x_k = \left(\frac{q}{p}\right)^k (x_1 - x_0).$$

Now,

$$\begin{aligned} x_k - x_0 &= (x_1 - x_0) + (x_2 - x_1) + \cdots + (x_k - x_{k-1}) \\ &= (x_1 - x_0)\left[1 + \frac{q}{p} + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{k-1}\right] \end{aligned} \tag{8}$$

So for $p \neq q$, we have

$$x_k - x_0 = (x_1 - x_0)\left(\frac{1-(q/p)^k}{1-(q/p)}\right), \quad q \neq p$$

Now using the boundary conditions $x_0 = 0$ and $x_N = 1$,

$$x_N - x_0 = 1 = (x_1 - x_0)\left(\frac{1-(q/p)^N}{1-(q/p)}\right)$$

and so by combining the two above equations and noting that $x_0 = 0$, we have shown the following result.

Theorem 4.2. *For $p \neq q$ and all $0 \leq k \leq N$,*

$$\mathbb{P}_k(T_N < T_0) = x_k = \frac{1-(q/p)^k}{1-(q/p)^N}.$$

For $p = q = 1/2$, we have that

$$\mathbb{P}_k(T_N < T_0) = \frac{k}{N}$$

The second result follows by going back to the summation in (8) or it follows from the first one by sending $p \downarrow q$.

4.3 Gambler's Ruin - Expected duration of the game

Given any event $B \in \mathcal{F}$ and a random variable X, we will use the following notation: The expectation of X and B is defined by

$$\mathbb{E}(X;B) := \mathbb{E}(X\mathbb{1}_B) =: \int_B X(\omega)\,\mathrm{d}\mathbb{P}(\omega)$$

With this notation we can define the conditional expectation.

Definition 4.3. For any $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, the *conditional expectation* of X given B is given by

$$\mathbb{E}(X|B) = \frac{\mathbb{E}(X;B)}{\mathbb{P}(B)} = \frac{\mathbb{E}(X\mathbb{1}_B)}{\mathbb{P}(B)}$$

Then, one can show that it is possible to split $\mathbb{E}(X)$ relative to B, i.e. we have that

$$\mathbb{E}(X) = \mathbb{E}\big(X(\mathbb{1}_B + \mathbb{1}_{B^c})\big) = \mathbb{P}(B)\,\mathbb{E}(X|B) + \mathbb{P}(B^c)\,\mathbb{E}(X|B^c). \tag{9}$$

In Gambler's ruin for SRW(p), let T be the random time when W first gets to 0 or N, so

$$T = T_0 \wedge T_N := \min\{T_0, T_N\}$$

Let $y_k := \mathbb{E}_k(T) = \mathbb{E}(T|W_0 = k)$. Then

$$\mathbb{E}_k(T) = \mathbb{P}_k(W_1 = k+1)\mathbb{E}_k(T|W_1 = k+1) + \mathbb{P}_k(W_1 = k-1)\mathbb{E}_k(T|W_1 = k-1)$$

so

$$y_k = p(1 + y_{k+1}) + q(1 + y_{k-1})$$

hence

$$y_k = py_{k+1} + qy_{k-1} + 1 \;\forall k \geq 1, \qquad \text{where } y_0 = 0,\, y_N = 0 \tag{10}$$

To solve this non-homogeneous difference equation, we first find a 'particular solution', that is, any solution of

$$z_k = pz_{k+1} + qz_{k-1} + 1,$$

where we ignore the boundary conditions. If we guess $z_k = ck$, we require

$$ck = pc(k+1) + qc(k-1) + 1 \Longrightarrow c = \frac{1}{q-p},$$

Hence, $z_k = \frac{k}{q-p}$. Secondly, we note that $w_k := y_k - z_k$ satisfies

$$w_k = pw_{k+1} + qw_{k-1} \text{ where } w_0 = 0,\, w_N = \frac{N}{p-q} \tag{11}$$

From difference equation theory, we look for solutions of the form w_k is proportional to λ^k, i.e. $w_k = a\lambda^k$ for some $a \in \mathbb{R}$. Substituting $w_k = a\lambda^k$ into (11) yields the following auxiliary equation

$$\lambda = p\lambda^2 + q \Longleftrightarrow p\lambda^2 - \lambda + q = 0 \Longleftrightarrow (\lambda - 1)(p\lambda - q) = 0$$

hence $\lambda = 1$ or $\lambda = q/p$. If $q \neq p$, then we have distinct roots for λ and the general solution is

$$w_k = A \times 1^k + B\left(\frac{q}{p}\right)^k \quad k = 0, \ldots, N$$

Now, if we take $y_k = w_k + z_k$, then y_k satisfies the recursion in (10) and we now have to choose A, B in order to satisfy the boundary conditions. Note that $z_0 = 0$ and $z_N = \frac{N}{q-p}$. Hence, in order to satisfy (10), we need that $y_0 = 0$ and $y_N = N$, which translates into $w_0 = 0$ and $w_N = -z_N = \frac{N}{p-q}$. Thus, A, B have to satisfy

$$A + B = 0 \text{ and } A + B\left(\frac{q}{p}\right)^N = \frac{N}{p-q}$$

Hence

$$w_k = \frac{N\left(1 - (q/p)^k\right)}{(p-q)\left(1 - (q/p)^N\right)} = \frac{Nx_k}{p-q}$$

Recalling that $\mathbb{E}_k(T) = y_k = w_k + z_k$, this suggests the following.

Theorem 4.4. *Let $p \neq \frac{1}{2}$. If T is the first time a SRW(p) hits 0 or N, then for $k \in \{1, \dots, N-1\}$,*

$$\mathbb{E}_k(T) = \frac{Nx_k - k}{p-q} = \frac{(N-k) - N(q/p)^k + k(q/p)^N}{(p-q)(1-(q/p)^N)}, \quad p \neq q$$

Proof. [Not examinable.] We have shown that $\mathbb{E}_k(T)$ satisfies the equation

$$y_k = 1 + py_{k+1} + qy_{k-1}, \quad y_0 = 0, y_N = N, \tag{12}$$

and we have found one solution (the right hand side). We still have to show that $\mathbb{E}_k(T)$ is the minimal non-negative solution of (12), thus we can exclude the trivial solution $y_k \equiv \infty$.

Let (y_k) be any non-negative solution of (12). Then,

$$y_k = 1 + py_{k+1} + qy_k = 1 + \sum_{\ell \notin \{0,N\}} \mathbb{P}_k(X_1 = \ell) y_\ell.$$

Iterating this equation, gives

$$\begin{aligned} y_k &= 1 + \sum_{\ell \notin \{0,N\}} \mathbb{P}_k(X_1 = \ell) + \sum_{\ell_1, \ell_2 \notin \{0,N\}} \mathbb{P}_k(X_1 = \ell_1)\mathbb{P}_{\ell_1}(X_1 = \ell_2) \\ &= 1 + \sum_{\ell \notin \{0,N\}} \mathbb{P}_k(X_1 = \ell) + \sum_{\ell_1, \ell_2 \notin \{0,N\}} \mathbb{P}_k(X_1 = \ell_1, X_2 = \ell_2) y_\ell, \end{aligned}$$

where we used the 'starting afresh' property. Reiterating again, we get

$$\begin{aligned} y_k &= 1 + \sum_{\ell \notin \{0,N\}} \mathbb{P}_k(X_1 = \ell) + \sum_{\ell_1, \ell_2 \notin \{0,N\}} \mathbb{P}_k(X_1 = \ell_1, X_2 = \ell_2) \\ &\quad + \sum_{\ell_1, \ell_2, \ell_3 \notin \{0,N\}} \mathbb{P}_k(X_1 = \ell_1, X_2 = \ell_2, X_3 = \ell_3) y_{\ell_3} \\ &\geq 1 + \sum_{\ell \notin \{0,N\}} \mathbb{P}_k(X_1 = \ell) + \sum_{\ell_1, \ell_2 \notin \{0,N\}} \mathbb{P}_k(X_1 = \ell_1, X_2 = \ell_2) + \dots \\ &\quad + \sum_{\ell_1, \dots, \ell_n \notin \{0,N\}} \mathbb{P}_k(X_i = \ell_i, i = 1, \dots, n) \\ &= \mathbb{P}_k(T \geq 1) + \mathbb{P}_k(T \geq 2) + \dots + \mathbb{P}_k(T \geq n) \end{aligned}$$

where we used that $y_k \geq 0$. Letting $n \to \infty$, we have seen that

$$y_k \geq \sum_{\ell \geq 1} \mathbb{P}_k(T \geq \ell) = \int_0^\infty \mathbb{P}_k(T \geq t)\, dt = \mathbb{E}_k(T),$$

by the identity proved on the problem sheet. $\square$

5 Generating Functions

In this section, we introduce the tool of generating functions, which will be one of the main techniques used in this unit. These will allows us to determine the distribution of random variables and in particular, we will use it to determine limit laws of a sequence of random variables.

Definition 5.1. For a random variable X taking values in $\mathbb{Z}^+ \cup \{\infty\}$, we define the *probability generating function* (PGF) g_X, of X as $g_X : [0,1) \to [0,1]$ with

$$g_X(\theta) := \mathbb{E}(\theta^X) = \sum_{k=0}^{\infty} \theta^k \, \mathbb{P}(X = k)$$

Here, we use the convention that $\theta^\infty = 0$ for $0 \le \theta < 1$.

PGFs have radius of convergence of at least 1. In particular, for $|\theta| < 1$, we can swap limits and differentiation and integration with the infinite sum.

Examples 5.2.

(i) For the Bernoulli distribution $X \sim \mathrm{Ber}(p)$, we have that

$$g_X(\theta) = \theta^0 \mathbb{P}(X = 0) + \theta \mathbb{P}(X = 1) = (1 - p) + p\theta.$$

(ii) For the Poisson distribution $Y \sim Po(\lambda)$, we have that

$$g_Y(\theta) = \sum_{k=0}^{\infty} \theta^k e^{-\lambda} \frac{\lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\theta\lambda)^k}{k!} = e^{\lambda(\theta-1)}.$$

Remark 5.3. We have the following

$$g_X(1-) := \lim_{\theta \uparrow 1} g_X(\theta) = \mathbb{P}(X < \infty)$$

by MON since $\theta^X \uparrow \mathbb{1}_{\{X<\infty\}}$ for $\theta \uparrow 1$ (since $\theta^X = 0$ on the event $\{X = \infty\}$). Moreover, note that for $0 \le \theta < 1$,

$$g'_X(\theta) = \mathbb{E}(X\theta^{X-1}).$$

and hence

$$g'_X(1-) = \mathbb{E}(X; X < \infty).$$

The arguments in the previous remark show the following result:

Lemma 5.4. *If* $\mathbb{P}(X < \infty) = 1$, *then*

$$\begin{cases} g_X(1) = 1, \; g'_X(1) = \mathbb{E}(X) \\ g''_X(1) = \mathbb{E}(X(X-1)) \\ g_X^{(k)}(1) = \mathbb{E}(X(X-1)\cdots(X-k+1)) \end{cases}$$

One reason why generating functions are so useful is that they interact well with sums of independent random variables.

Theorem 5.5. *If X and Y are independent random variables with values in $\mathbb{Z}^+ \cup \{\infty\}$, then*

$$g_{X+Y}(\theta) = g_X(\theta) g_Y(\theta)$$

Proof. If X and Y are independent, then so are θ^X and θ^Y, so

$$\mathbb{E}(\theta^{X+Y}) = \mathbb{E}(\theta^X \theta^Y) = \mathbb{E}(\theta^X)\mathbb{E}(\theta^Y)$$

by independence, as required. □

Examples 5.6.

(i) Binomial sums; if $X_n \sim B(n,p)$ then $X_n = Y_1 + \cdots + Y_n$ where $Y_1, Y_2, \ldots$ are IID, $Y_i \sim B(1,p)$ with $\mathbb{P}(Y_i = +1) = p$, $\mathbb{P}(Y_i = -1) = 1 - p$ and

$$g_{Y_i}(\theta) = p\theta + (1-p)1 = p\theta + (1-p)$$

hence

$$g_{X_n}(\theta) = \mathbb{E}\left(\theta^{Y_1 + \cdots + Y_n}\right) = \prod_{i=1}^{n} \mathbb{E}\left(\theta^{Y_i}\right) = (q + p\theta)^n$$

(ii) If $X \sim \mathrm{Po}(\lambda)$ and $Y \sim \mathrm{Po}(\mu)$ with X and Y independent, then

$$g_{X+Y}(\theta) = g_X(\theta) g_Y(\theta) = e^{\lambda(\theta-1)} e^{\mu(\theta-1)} = e^{(\lambda+\mu)(\theta-1)}$$

This is the PGF of a $\mathrm{Po}(\lambda+\mu)$ random variable, hence $X + Y \sim \mathrm{Po}(\lambda+\mu)$ by the following theorem.

Theorem 5.7. *The PGF g_Z uniquely determines the distribution of the random variable Z. In particular,*

$$\mathbb{P}(Z = k) = \frac{g_Z^{(k)}(0)}{k!} = \frac{1}{k!} \frac{\mathrm{d}^k}{\mathrm{d}\theta^k} g_Z(\theta)\bigg|_{\theta=0}$$

and

$$\mathbb{P}(Z = +\infty) = 1 - \sum_{k=1}^{\infty} \mathbb{P}(Z = k) = 1 - \mathbb{P}(Z < \infty) = 1 - g_Z(1-)$$

In practice, if the PGF is recognisable, then we already have the distribution. Otherwise, if we can expand a PGF $g_Z(\theta)$ as a power series expansion in θ, that is

$$g_Z(\theta) = \sum_{i=0}^{\infty} p_i \theta^i$$

for some $\{p_i : i \in \mathbb{Z}^+\}$, then necessarily $\mathbb{P}(Z = i) = p_i$.

Sometimes, the following identity can be useful when multiplying power series:

$$\Big(\sum_{k=0}^{\infty} a_k\theta^k\Big)\Big(\sum_{\ell=0}^{\infty} b_\ell\theta^\ell\Big) = \sum_{n=0}^{\infty} c_n\theta^n, \tag{13}$$

where

$$c_n = \sum_{m=0}^{n} a_m b_{n-m}.$$

[To check, multiply out the LHS and equate coefficients of θ^n.]

Theorem 5.8 (Convergence of PGFs). *Let $X, X_1, X_2, X_3, \dots$ be a sequence of random variables on $\mathbb{Z}^+ \cup \{\infty\}$. Then the following are equivalent:*

(i) The PGFs g_{X_n} converge pointwise to the PGF g_X as $n \to \infty$, that is, for all $\theta \in [0,1)$,

$$\lim_{n\to\infty} g_{X_n}(\theta) = g_X(\theta)$$

(ii) The distributions of X_n for $n \geq 1$ converge to the distribution of X, that is, for all $k \geq 0$,

$$\lim_{n\to\infty} \mathbb{P}(X_n = k) = \mathbb{P}(X = k)$$

We say that $(X_n)_{n\geq 1}$ converges in distribution to X, written $X_n \xrightarrow{\mathcal{D}} X$, or $(X_n)_{n\geq 1}$ converges in law to X, $X_n \xrightarrow{\mathcal{L}} X$.

Proof. Suppose that $g_{X_n}(\theta) \to g_X(\theta)$ $\forall\theta \in [0,1)$.

$$\begin{aligned}
\sum_{k=0}^{\infty} \theta^k \mathbb{P}(X = k) = g_X(\theta) &= \lim_{n\to\infty} g_{X_n}(\theta) \\
&= \lim_{n\to\infty} \sum_{k=0}^{\infty} \theta^k \mathbb{P}(X_n = k) \\
&= \sum_{k=0}^{\infty} \lim_{n\to\infty} \theta^k \mathbb{P}(X_n = k) \\
&= \sum_{k=0}^{\infty} \theta^k \lim_{n\to\infty} \mathbb{P}(X_n = k)
\end{aligned}$$

since the PGF is uniformly convergent for $\theta \in [0,1)$. Then, as the power series are unique, comparing θ^k coefficients gives

$$\mathbb{P}(X = k) = \lim_{n\to\infty} \mathbb{P}(X_n = k)$$

This proves that (i) $\implies$ (ii). For the backwards implication, we just reverse the argument. $\square$

Example 5.9 (Poisson approximation of binomial random variables)**.** Let $X_n \sim B(n, p_n)$ where $p_n = \lambda/n$. Then

$$g_{X_n}(\theta) = (1 - p_n + p_n\theta)^n = \left(1 - \frac{\lambda}{n} + \frac{\lambda}{n}\theta\right)^n = \left(1 + (\theta - 1)\frac{\lambda}{n}\right)^n \to e^{(\theta-1)\lambda},$$

as $n \to \infty$. But if $X \sim \mathrm{Po}(\lambda)$, then $g_X(\theta) = e^{(\theta-1)\lambda}$, so $g_{X_n}(\theta) \to g_X(\theta)$ $\forall \theta \in [0, 1)$. Hence $X_n \xrightarrow{\mathcal{D}} X$, that is, for all $k \in \mathbf{Z}_+$

$$\lim_{n\to\infty} \mathbb{P}(X_n = k) = \mathbb{P}(X = k) = \frac{\lambda^k}{k!}e^{-\lambda}$$

In other words, for n large, X_n is approximately $\mathrm{Po}(\lambda)$ distributed.

6 More on random walks

In this chapter, we will come back to analysing simple random walks. In the first part, we will apply the technique of generating functions to describe the distribution of the first return time to the origin. In the second part of the chapter, we will introduce the reflection principle, which allows us to compute further quantities related to the simple random walk, such as the distribution of its maximum.

6.1 Distribution of the return time for a simple random walk

Let $W = (W_n)_{n\geq 0}$ be a SRW(p). Define the return time to r as

$$H_r := \inf\{k > 0 : W_k = r\}.$$

Note that $H_r > 0$ and $H_r = +\infty$ if $W_k \neq r\ \forall k > 0$. Let g_r be the PGF of H_r given $W_0 = 0$:

$$g_r(\theta) := \mathbb{E}_0(\theta^{H_r}) = \mathbb{E}(\theta^{H_r}|W_0 = 0)$$

Theorem 6.1. *(i) For any $r > 0$,*

$$g_r(\theta) = g_1(\theta)^r, \;\; g_1(\theta) = \frac{1 - \sqrt{1-4pq\theta^2}}{2q\theta}$$

$$g_{-r}(\theta) = g_{-1}(\theta)^r, \;\; g_{-1}(\theta) = \frac{1 - \sqrt{1-4pq\theta^2}}{2p\theta}$$

For $r = 0$, i.e. for the return time H_0 to 0, we have

$$g_0(\theta) = \mathbb{E}(\theta^{H_0}) = 1 - \sqrt{1-4pq\theta^2}.$$

(ii) Moreover,

$$\begin{aligned}
\mathbb{P}_0(H_0 = \infty) &= \mathbb{P}_0(\textit{no return}) = |p-q| \\
\mathbb{P}_0(H_0 = 2n) &= \frac{1}{2n-1}\binom{2n}{n}p^n q^n \;(n \geq 1) \\
\mathbb{E}_0(H_0|H_0 < \infty) &= \frac{1+|p-q|}{|p-q|},\; \mathbb{E}_0(H_0) = +\infty
\end{aligned}$$

Note for $p = q = 1/2$, $\mathbb{P}_0(H_0 < \infty) = 1$ but $\mathbb{E}_0(H_0) = +\infty$.

Proof. (i) Let $\theta \in [0, 1)$. By splitting over the time to hit 1, we see that

$$g_r(\theta) = \mathbb{E}_0(\theta^{H_r}) = \mathbb{E}_0(\theta^{H_r}; H_1 < \infty)$$

since $H_r \geq H_1$ and $\theta^{H_r} \leq \theta^{H_1} = 0$ if $H_1 = +\infty$ and $\theta \in [0,1)$. So

$$\begin{aligned} g_r(\theta) &= \sum_{k=1}^{\infty} \mathbb{E}_0(\theta^{H_r}; H_1 = k) \\ &= \sum_{k=1}^{\infty} \mathbb{P}_0(H_1 = k)\mathbb{E}_0(\theta^{H_r}|H_1 = k) \end{aligned}$$

Now, the increments $X_{k+1}, X_{k+2}, \ldots$ are independent from $W_0, \ldots, W_k$. Hence, we can continue

$$\begin{aligned} g_r(\theta) &= \sum_{k=1}^{\infty} \mathbb{P}_0(H_1 = k)\, \theta^k\, \mathbb{E}_1(\theta^{H_r}) \\ &= g_{r-1}(\theta) \sum_{k=1}^{\infty} \theta^k\, \mathbb{P}_0(H_1 = k) \\ &= g_{r-1}(\theta) g_1(\theta) \end{aligned}$$

Iterating the arguments yields,

$$g_r(\theta) = g_1(\theta)^r,\ r \geq 1 \text{ and } g_{-r}(\theta) = g_{-1}(\theta)^r,\ r \geq 1$$

Now if we split over the first move, we see that

$$\begin{aligned} g_1(\theta) = \mathbb{E}_0(\theta^{H_1}) &= p\mathbb{E}_0(\theta^{H_1}|W_1 = 1) + (1-p)\mathbb{E}_0(\theta^{H_1}|W_1 = -1) \\ &= p\theta + (1-p)\theta\mathbb{E}_{-1}(\theta^{H_1}) \end{aligned}$$

where

$$\mathbb{E}_{-1}(\theta^{H_1}) = \mathbb{E}_0(\theta^{H_2}) = g_2(\theta) = g_1(\theta)^2$$

This implies that

$$g_1(\theta) = p\theta + (1-p)\theta g_1(\theta)^2 \Longrightarrow q\theta g_1(\theta)^2 - g_1(\theta) + p\theta = 0$$

so $g_1(\theta)$ must be one of the roots

$$\frac{1\mathbb{P}m\sqrt{1-4pq\theta^2}}{2q\theta}$$

but $g_1(\theta) = \mathbb{E}_0(\theta^{H_1}) \in [0,1)\ \forall \theta \in [0,1)$, whereas

$$\frac{1+\sqrt{1-4pq\theta^2}}{2q\theta} \to +\infty \text{ as } \theta \to 0$$

hence

$$g_1(\theta) = \frac{1-\sqrt{1-4pq\theta^2}}{2q\theta} \tag{14}$$

and similarly (by symmetry),

$$g_{-1}(\theta) = \frac{1 - \sqrt{1 - 4pq\theta^2}}{2p\theta} \tag{15}$$

It remains to consider $g_0(\theta)$. Again, by splitting according to the first move,

$$g_0(\theta) = \mathbb{E}_0(\theta^{H_0}) = p\theta\mathbb{E}_1(\theta^{H_0}) + q\theta\mathbb{E}_{-1}(\theta^{H_0}) \qquad = p\theta g_{-1}(\theta) + q\theta g_1(\theta)$$

Hence, by (14) and (15),

$$g_0(\theta) = 1 - \sqrt{1 - 4pq\theta^2}$$

Thus, we have proved part (i) of the theorem.

(ii) By Remark 5.3

$$\begin{aligned}
\mathbb{P}_0(H_0 < \infty) = g_0(1-) &:= \lim_{\theta \uparrow 1} g_0(\theta) \\
&= 1 - \sqrt{1 - 4pq} \\
&= 1 - \sqrt{(2p-1)^2} = 1 - \sqrt{(p-q)^2} \\
&= 1 - |p - q|.
\end{aligned}$$

Hence,

$$\mathbb{P}_0(H_0 = +\infty) = |p - q|$$

which is the probability of no return to the origin.

Also

$$g_0'(\theta) = \mathbb{E}_0(H_0\theta^{H_0 - 1}; H_0 < \infty) = \frac{\mathbb{P}artial}{\mathbb{P}artial\theta}\left(1 - \sqrt{1 - 4pq\theta^2}\right) = \frac{4pq\theta}{\sqrt{1 - 4pq\theta^2}}$$

and then

$$\mathbb{E}_0(H_0; H_0 < \infty) = g_0'(1-) := \lim_{\theta \uparrow 1} \frac{4pq\theta}{\sqrt{1 - 4pq\theta^2}}$$

$$= \begin{cases} \dfrac{4pq}{\sqrt{1 - 4pq}} = \dfrac{4pq}{|p - q|}, & p \neq q \\ \qquad +\infty, & p = q = \frac{1}{2} \end{cases}$$

Hence, for $p \neq q$,

$$\begin{aligned}
\mathbb{E}_0(H_0 | H_0 < \infty) = \frac{\mathbb{E}_0(H_0; H_0 < \infty)}{\mathbb{P}_0(H_0 < \infty)} &= \frac{4pq}{|p - q|}\left(\frac{1}{1 - \sqrt{1 - 4pq}}\right) \\
&= \frac{4pq(1 + \sqrt{1 - 4pq})}{|p - q|\big(1 - (1 - 4pq)\big)} = \frac{1 + |p - q|}{|p - q|}
\end{aligned}$$

Note that since $\mathbb{P}_0(H_0 = +\infty) > 0$ for $p \neq q$, it follows that $\mathbb{E}_0(H_0) = +\infty$. If $p = q = 1/2$, then $\mathbb{P}_0(H_0 = \infty) = 0$, so $\mathbb{E}_0(H_0) = \mathbb{E}(H_0; H_0 < \infty) = +\infty$ also.

Finally to recover the distribution we note that by a series expansion

$$1-\sqrt{1-x}=\sum_{n=1}^{\infty}\binom{2n}{n}\frac{x^n}{2^{2n}}\frac{1}{2n-1}\quad\forall x\in[0,1)$$

hence,

$$\begin{aligned}g_0(\theta)&=1-\sqrt{1-4pq\theta^2}\\&=\sum_{n=1}^{\infty}\binom{2n}{n}\frac{p^nq^n}{2n-1}\theta^{2n}\end{aligned}$$

but

$$g_0(\theta):=\mathbb{E}_0(\theta^{H_0})=\sum_{k=0}^{\infty}\theta^k\mathbb{P}_0(H_0=k)$$

by definition, so by equating coefficients of θ^k we obtain that

$$\mathbb{P}_0(H_0=2n)=\binom{2n}{n}\frac{p^nq^n}{2n-1}\quad\forall n\in\mathbb{N}$$

and

$$\mathbb{P}_0(H_0=2n-1)=0\;\;\forall n\in\mathbb{N}.\qquad\square$$

6.2 The Reflection Principle

Let $m, n, x, y \in \mathbb{Z}$ with $n > m$. How many paths go from (m, x) to (n, y)?

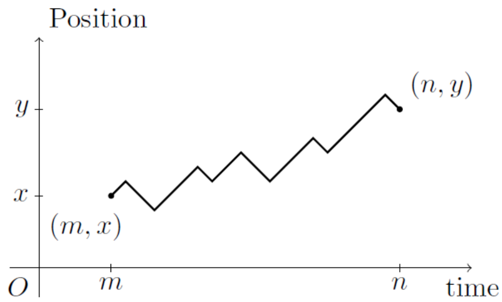


Figure 4: The reflection principle.

Let u be the number of 'ups', i.e. the number of $+1$ jumps and let d be the number of 'downs', i.e. the number of -1 jumps. Then we need

$$\begin{cases} u + d = n - m, & \text{total time for the path} \\ u - d = y - x, & \text{change in position} \end{cases}$$

Hence

$$u = \frac{(n-m)+(y-x)}{2} \quad \text{and} \quad d = \frac{(n-m)-(y-x)}{2} = (n-m) - u \tag{16}$$

The 'ups' can occur in any u of the $n - m$ time intervals, hence the number of paths from (m, x) to (n, y) is

$$\binom{n-m}{u} \quad \text{where } u = \frac{(n-m)+(y-x)}{2} \tag{17}$$

with the convention that the Binomial coefficient is 0 unless $u \in \{0, 1, \ldots, n - m\}$.

Remark 6.2. If there is a path from (m, x) to (n, y) and $n - m$ is even (or respectively odd), then $y - x$ must also be even (odd). Clearly we need both u and d to be integers in the range $\{0, 1, \ldots, n - m\}$ or no path can go from (m, x) to (n, y).

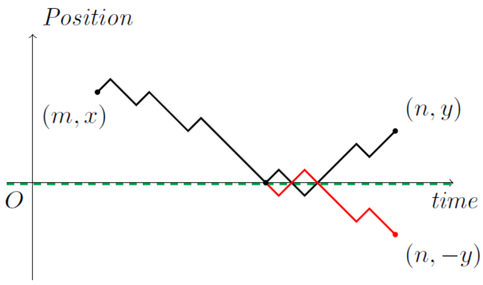
Corollary 6.3. *For SRW(p),*

$$\mathbb{P}_0(W_n = 2u - n) = \binom{n}{u} p^u q^{n-u} \quad \textit{for } u \in \{0, 1, \ldots, n\}$$

Note that $W_n = 2u - n$ corresponds to u 'ups' and $n - u$ 'downs' when $W_0 = 0$ (by (16) with $x = 0, m = 0$). If $W_0 = 0$, then W_n can only be at positions $\{-n, -n + 2, \ldots, n - 2, n\}$.

Theorem 6.4 (Reflection principle for paths)**.** *Let $x, y > 0$ and $n > m$. Then*

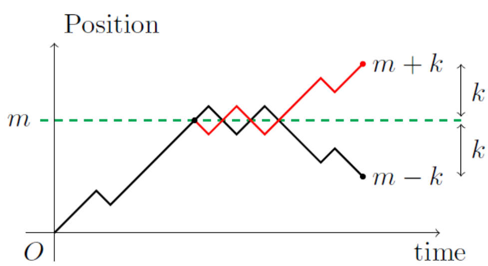
$$\#\left\{\begin{array}{c}\textit{paths from } (m,x) \textit{ to } (n,y) \\ \textit{which touch or cross the} \\ \textit{time axis, } \{(k,0) : k \in \mathbb{Z}\}\end{array}\right\} = \#\left\{\begin{array}{c}\textit{paths from } (m,x) \\ \textit{to } (n,-y)\end{array}\right\} = \#\left\{\begin{array}{c}\textit{paths from } (m,-x) \\ \textit{to } (n,y)\end{array}\right\}$$



Proof. Any path that reaches the time-axis can be 'reflected' from the **first** time it hits to give a path ending at $(n, -y)$. This gives a one-to-one correspondence between paths touching the time-axis ending at (n, y) and those paths ending at $(n, -y)$. ☐

Theorem 6.5 (Reflection principle for symmetric simple RW)**.** *Let $M_n := \max\{W_k : k \leq n\}$ (the maximum of the random walk) for $W = (W_n)_{n\geq 0}$ a SRW$(1/2)$, then $\forall k, m \in \mathbb{Z}^+$,*

$$\mathbb{P}_0(M_n \geq m; W_n = m - k) = \mathbb{P}_0(W_n = m + k)$$



Proof. By the Reflection Principle, there is a one-to-one correspondence of paths reaching

m and ending at $m-k$ and paths ending at $m+k$, so that

$$\begin{aligned}
\mathbb{P}_0(M_n \geq m; W_n = m-k) &= \left(\frac{1}{2}\right)^n \# \left\{ \begin{array}{l} \text{paths such that} \\ M_n \geq m,\ W_n = m-k \end{array} \right\} \\
&= \left(\frac{1}{2}\right)^n \# \left\{ \begin{array}{l} \text{paths with} \\ W_n = m+k \end{array} \right\} \\
&= \mathbb{P}_0(W_n = m+k)
\end{aligned}$$

□

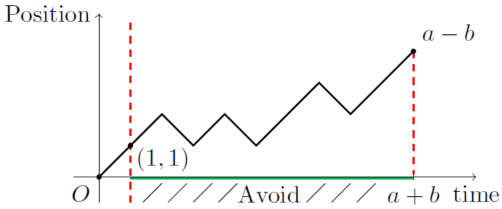
Theorem 6.6 (Ballot theorem)**.** *Consider an election between two candidates A and B where A gets a votes and B gets b votes with $a > b$. If slips are counted in a random order, then the probability that A is strictly in the lead from the first slip counted until the end of the count is*

$$\frac{a-b}{a+b}$$

Proof. Each count corresponds to a path from $(0,0)$ to $(a+b, a-b)$ and each path is equally likely as the slips are in a random order (the number of ups is a, and number of downs b). Then

$$\#\{\text{paths from } (0,0) \text{ to } (a+b, a-b)\} = \binom{a+b}{a}$$

For A to always be strictly in the lead, the first slip must be for A (an 'up') and then the path must not touch 0 again.



Note that

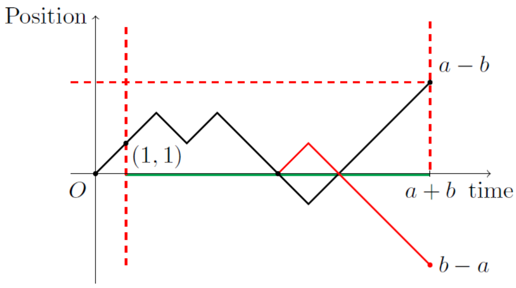
$$\#\{\text{paths from } (1,1) \text{ to } (a+b, a-b)\} = \binom{a+b-1}{b}$$

but by the reflection principle

$$\begin{aligned}
\# \left\{ \begin{array}{c} \text{paths from } (1,1) \text{ to } (a+b, a-b) \\ \text{that touch the time axis} \end{array} \right\} &= \# \left\{ \begin{array}{c} \text{paths from } (1,1) \text{ to} \\ (a+b, -(a-b)) \end{array} \right\} \\
&= \binom{a+b-1}{b-1},
\end{aligned}$$

and where for the second equality we use that the number of 'up' steps required in the path from $(1,1)$ to $(a+b, -(a-b))$ is given as in (17) by

$$u := \frac{1}{2}(a+b-1+(-(a-b)-1)) = b-1.$$



Then,

$$
\begin{aligned}
&\#\left\{\begin{array}{c}\text{paths } (1,1) \text{ to } (a+b, a-b)\\ \text{avoiding hitting } 0\end{array}\right\}\\
&\quad = \#\left\{\begin{array}{c}\text{paths } (1,1) \text{ to}\\ (a+b, a-b)\end{array}\right\} - \#\left\{\begin{array}{c}\text{paths } (1,1) \text{ to } (a+b, a-b)\\ \text{that hit } 0 \text{ line}\end{array}\right\}\\
&\quad = \binom{a+b-1}{b} - \binom{a+b-1}{b-1}
\end{aligned}
$$

Hence, as all paths are equally likely,

$$
\begin{aligned}
\mathbb{P}\left(\begin{array}{c}A \text{ always strictly}\\ \text{in the lead}\end{array}\right) &= \frac{\binom{a+b-1}{b} - \binom{a+b-1}{b-1}}{\binom{a+b}{b}}\\
&= \frac{a!b!}{(a+b)!}\left\{\frac{(a+b-1)!}{b!(a-1)!} - \frac{(a+b-1)!}{(b-1)!a!}\right\}\\
&= \frac{1}{a+b}(a-b) = \frac{a-b}{a+b}
\end{aligned}
$$

□

Corollary 6.7. *For SRW(p),*

$$\mathbb{P}_0(W_k \neq 0,\, 1 \leq k \leq 2n-1 \,|\, W_{2n-1} = 1) = \frac{1}{2n-1}$$

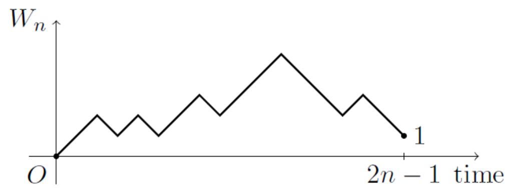
Proof. All given paths of SRW(p) that end at 1 at time $2n-1$ are equally likely with probability $p^n q^{n-1}$ as each path has n ups and $n-1$ downs chosen at random from $2n-1$ time slots.

The Ballot theorem then tells us the required probability is

$$\frac{n-(n-1)}{n+(n-1)} = \frac{1}{(2n-1)}.$$

□

Using the reflection principle, we can give an alternative proof of Theorem 6.1 describing the distribution of the first return time to 0 for the random walk.



Theorem 6.8 (Application to return times)**.** *For a SRW(p) started at 0, let H_0 be the first time the particle returns to 0. Then,*

$$\mathbb{P}_0(H_0 = 2n) = \frac{2}{2n-1}\binom{2n-1}{n}p^n q^n$$

Proof. Splitting into paths that stay above, resp. below, the 0-line and then using Corollary 6.7

$$\begin{aligned}
\mathbb{P}_0(H_0 = 2n) &= \mathbb{P}_0(W_{2n} = 0; W_k \neq 0 \text{ for } k = 1, \ldots, 2n-1) \\
&= \mathbb{P}(W_{2n} = 0; W_{2n-1} = +1; W_k \neq 0 \text{ for } k = 1, \ldots, 2n-1) \\
&\quad + \mathbb{P}_0(W_{2n} = 0; W_{2n-1} = -1; W_k \neq 0 \text{ for } k = 1, \ldots, 2n-1) \\
&= \mathbb{P}_0(W_{2n-1} = +1)\mathbb{P}_0(W_k \neq 0,\ \forall k \,|\, W_{2n-1} = +1) \\
&\qquad \times \mathbb{P}_0(W_{2n} = 0 \,|\, W_{2n-1} = +1, W_k \neq 0, \forall k) \\
&\quad + \mathbb{P}_0(W_{2n-1} = -1)\mathbb{P}_0(W_k \neq 0,\ \forall k \,|\, W_{2n-1} = -1) \\
&\qquad \times \mathbb{P}_0(W_{2n} = 0 \,|\, W_{2n-1} = -1, W_k \neq 0, \forall k) \\
&= \binom{2n-1}{n}p^n q^{n-1} \cdot \frac{1}{2n-1} \cdot q + \binom{2n-1}{n}p^{n-1}q^n \cdot \frac{1}{2n-1} \cdot p \\
&= \frac{2}{2n-1}\binom{2n-1}{n}p^n q^n.
\end{aligned}$$

□

This answer agrees with Theorem 6.1 since

$$\binom{2n}{n} = \frac{2n(2n-1)!}{n \times (n-1)! \times n!} = 2\binom{2n-1}{n}.$$

7 Weak and Strong Laws

The goal of this chapter is to introduce the *law of large numbers*, which states that if $X_1, X_2, \dots, X_n$ are i.i.d. random variables with expectation $\mu = \mathbb{E}(X_1)$, then

$$\frac{1}{n}\sum_{i=1}^{n} X_i \to \mu.$$

The statement is probably not surprising. Think e.g. of tossing a coin repeatedly and taking $X_i = \mathbb{1}_{\{H\}}$. Then, you expect that $\frac{1}{n}\sum_{i=1}^{n} X_i$, i.e. the proportion of coin tosses that land on head, is close to $\mathbb{P}(H) = \mathbb{E}(\mathbb{1}_{\{H\}})$. In the following, we will make the meaning of '$\to$' more precise. In particular, we will see in this and also the next chapter that in probability there are various different types of convergence.

Before we can state the first version of the law of large numbers, we need the following inequality:

Lemma 7.1 (Chebyshev's inequality)**.** *If $X \in \mathcal{L}^2$, that is $\mathbb{E}(X^2) < \infty$, and $c > 0$, then for $\mu := \mathbb{E}(X)$,*

$$\mathbb{P}\big(|X - \mu| \geq c\big) \leq \frac{\mathrm{Var}(X)}{c^2}.$$

We recall that $\mathrm{Var}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$.

Proof. Note that for all $\mu \in \mathbb{R}$ and $c > 0$,

$$c^2 \mathbb{1}_{\{|X-\mu|>c\}} = c^2 \mathbb{1}_{\{|X-\mu|^2>c^2\}} \leq (X - \mu)^2.$$

Now, letting $\mu := \mathbb{E}(X)$ and taking expectations, we obtain

$$\mathbb{E}\big(c^2 \mathbb{1}\{|X - \mu| \geq c\}\big) \leq \mathbb{E}\big((X - \mu)^2\big).$$

Therefore,

$$c^2 \mathbb{P}\big(|X - \mu| \geq c\big) \leq \mathrm{Var}(X).$$

Rearranging gives the statement of the theorem. □

Remark 7.2. Note that $\mathbb{E}(X^2) < \infty$ implies that $\mathbb{E}|X| < \infty$, so $X \in \mathcal{L}^1$. Hence $\mathbb{E}(X) < \infty$ and $\mathrm{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 < \infty$.

Theorem 7.3 (Weak law of large numbers, WLLN)**.** *Let $X_1, X_2, \dots$ be IID random variables such that $\mathbb{E}(X_1^2) < \infty$ and common expectation $\mu := \mathbb{E}(X_1)$. Then the* weak law of large numbers *holds in that, for any $\varepsilon > 0$,*

$$\mathbb{P}\left(\left|\frac{\sum_{i=1}^{n} X_i}{n} - \mu\right| > \varepsilon\right) \to 0 \text{ as } n \to \infty$$

Remark 7.4. The conditions we give above on $X_1, X_2, \ldots$ are not optimal. In fact, it is possible to show that the WLLN holds if and only if one of the following conditions holds:

(i) $n\mathbb{P}\big(|X_1| > n\big) \to 0$ and $\mathbb{E}(X_1; |X_1| \le n) \to \mu$ as $n \to \infty$;

(ii) The characteristic function of X_1, $\varphi(t) := \mathbb{E}(e^{itX_1})$, is differentiable at $t = 0$ and $\varphi'(0) = i\mu$.

See also [GS01, Section 7.4]

Proof. Let $S_n := \sum_{i=1}^n X_i$. Then

$$\mathbb{E}(S_n) = \mathbb{E}\Big(\sum_{i=1}^n X_i\Big) = \sum_{i=1}^n \mathbb{E}(X_i) = n\mu$$

and by independence

$$\operatorname{Var}(S_n) = \operatorname{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \operatorname{Var}(X_i) = n\operatorname{Var}(X_1)$$

Now $\sigma^2 := \operatorname{Var}(X_1) < \infty$ since $X_1 \in \mathcal{L}^2$. Also,

$$\mathbb{E}(S_n/n) = \frac{1}{n}\mathbb{E}(S_n) = \mu$$

and

$$\operatorname{Var}(S_n/n) = \frac{1}{n^2}\operatorname{Var}(S_n) = \frac{\sigma^2}{n}$$

Therefore, for any $\varepsilon > 0$, by Chebyshev's inequality,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \varepsilon\right) \le \frac{1}{\varepsilon^2}\operatorname{Var}\left(\frac{S_n}{n}\right) = \frac{\sigma^2}{\varepsilon n} \to 0,$$

as $n \to \infty$. $\square$

Theorem 7.5 (Strong law of large numbers, SLLN)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables. Then,*

$$\mathbb{P}\left(\frac{\sum_{i=1}^n X_i}{n} \to \mu\right) = 1, \tag{18}$$

for some constant μ if and only if $\mathbb{E}|X_1| < \infty$. In this case $\mu = \mathbb{E}(X_1)$.

Note that

$$\begin{aligned}\Big\{\frac{\sum_{i=1}^n X_i}{n} \to \mu\Big\} &= \Big\{\omega \in \Omega : \frac{\sum_{i=1}^n X_i(\omega)}{n} \to \mu\Big\}\\ &= \left\{\omega \in \Omega : \forall\varepsilon > 0, \exists N(\omega) \in \mathbb{N} \text{ s.t. } \left|\frac{\sum_{i=1}^n X_i(\omega)}{n} - \mu\right| \le \varepsilon,\ \forall n \ge N(\omega)\right\}\end{aligned}$$

In particular, N depends on both the choice of $\omega \in \Omega$ and $\varepsilon > 0$, in general.

Proof. We will only prove the following special case: We suppose additionally that $\mathbb{E}(X_1^2) < \infty$. For the general case see [GS01, Section 7.5].

We will first show that $\frac{1}{n}S_n$ converges along the subsequence $n_i = i^2$ and then 'fill the gaps' to recover the full convergence.

By Chebyshev's inequality, Lemma 7.1, we have that (using similar calculations as in the previous theorem),

$$\mathbb{P}\Big(\Big|\frac{1}{i^2}S_{i^2} - \mu\Big| > \varepsilon\Big) \leq \frac{\mathrm{Var}(S_{i^2})}{i^4\varepsilon} = \frac{\mathrm{Var}(X_1)}{i^2\varepsilon^2}.$$

The right hand side is summable in i, so that by the Borel-Cantelli Lemma I, only finitely many of the events on the right hand side occur. In particular, for any $\varepsilon > 0$, there exists $K = K(\omega, \varepsilon)$ such that for all $i \geq K$,

$$\Big|\frac{1}{i^2}S_{i^2} - \mu\Big| \leq \varepsilon.$$

Thus, $\mathbb{P}(\lim_{i\to\infty} \frac{1}{i^2}S_{i^2} = \mu) = 1$. To show the convergence of the full sequence, assume for the moment that the X_i are non-negative. In this case we have

$$S_{i^2} \leq S_n \leq S_{(i+1)^2}, \quad \text{for all } i^2 \leq n \leq (i+1)^2.$$

Dividing by n, we have

$$\frac{1}{(i+1)^2}S_{i^2} \leq \frac{1}{n}S_n \leq \frac{1}{i^2}S_{(i+1)^2}, \quad \text{for all } i^2 \leq n \leq (i+1)^2.$$

Now, note that $\frac{i}{i+1} \to 1$. Hence, we have shown that for non-negative random variables

$$\mathbb{P}\Big(\frac{1}{n}S_n \to \mu\Big) = 1. \tag{19}$$

Coming back to general (not necessarily non-negative) random variables, we decompose $X_i = X_i^+ - X_i^-$ into positive and negative part (recall the definitions from Section 2.2). X_i^+ and X_i^- are non-negative random variables, so that we can apply (19) and deduce that on a set of probability 1, we have that

$$\frac{1}{n}S_n = \frac{1}{n}\Big(\sum_{i=1}^n X_i^+ + \sum_{i=1}^n X_i^-\Big) \to \mathbb{E}(X_1^+) - \mathbb{E}(X_1^-) = \mathbb{E}(X_1),$$

as $n \to \infty$. □

Remark 7.6. Discussion. Let $S_n = \sum_{i=1}^n X_i$. The SLLN is stronger than the WLLN. The WLLN says that, for a given large enough n, $|\frac{1}{n}S_n - \mu|$ is likely to be small with a high probability. However, it does not imply that $|\frac{1}{n}S_n - \mu|$ remains small for all sufficiently large n with probability 1 – this is the SLLN. In fact,

$$\text{SLLN holds} \Longrightarrow \text{WLLN holds}$$

but

$$\text{WLLN holds} \not\Longrightarrow \text{SLLN holds}$$

in general.

For example, taking X_1 symmetric (so X_1 and $-X_1$ have the same distribution) and

$$\mathbb{P}(X_1 \geq x) \sim \frac{1}{x \log x}$$

for x large. Then $\mathbb{E}|X_1| = +\infty$, but $\mathbb{E}(X_1; |X_1| \leq n) = 0 \ \forall n$ by symmetry, so $\mu = 0$ and (i) holds in Remark 7.4. Hence, the WLLN holds, but the SLLN does not.

8 Modes of Convergence

There are four main ways to interpret $X_n \to X$ as $n \to \infty$.

Definition 8.1. Let $X, X_1, X_2, \dots$ be random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We say that;

(i) $X_n \to X$ *almost surely*, written $X_n \xrightarrow{\text{a.s.}} X$, if $\{\omega \in \Omega : X_n(\omega) \to X(\omega) \text{ as } n \to \infty\}$ is an event of probability 1. That is, $\mathbb{P}(X_n \to X) = 1$.

(ii) $X_n \to X$ in *$p-th$ mean*, where $p \geq 1$, written $X_n \xrightarrow{p^{\text{th}}} X$ or $X_n \to X$ in $\mathcal{L}^p$, if

$$\mathbb{E}\big(|X_n - X|^p\big) \to 0 \text{ as } n \to \infty$$

(iii) $X_n \to X$ in *probability*, written $X_n \xrightarrow{\text{prob.}} X$, if

$$\forall \varepsilon > 0,\ \mathbb{P}\big(|X_n - X| > \varepsilon\big) \to 0 \text{ as } n \to \infty$$

(iv) $X_n \to X$ in *distribution*, written $X_n \xrightarrow{\mathcal{D}} X$ or $X_n \xrightarrow{\text{Law}} X$, if

$$\mathbb{P}(X_n \leq x) \to \mathbb{P}(X \leq x) \text{ as } n \to \infty$$

for all points x at which $F_X(x) := \mathbb{P}(X \leq x)$ is continuous.

Remark 8.2. (i) There is a fundamental difference between the first three and the last type of convergence. For the first three, all random variables and the limit need to be defined on the same probability space, however, this does not apply to the convergence in distribution.

(ii) Note that with $S_n := X_1 + X_2 + \cdots + X_n$ and $\mu := \mathbb{E}(X)$, the SLLN says that $S_n/n \to \mu$ almost surely and the WLLN says that $S_n/n \to \mu$ in probability.

Theorem 8.3. *The following implications hold;*

$$\begin{array}{ll} X_n \to X \ \textit{a.s.} \ \searrow & \\ & X_n \to X \ \textit{in prob.} \Longrightarrow X_n \xrightarrow{\mathcal{D}} X \\ X_n \to X \ \textit{in } \mathcal{L}^p \ \nearrow & \end{array}$$

for any $p \geq 1$. Also, if $p > q \geq 1$ then

$$X_n \to X \ \textit{in } \mathcal{L}^p \Longrightarrow X_n \to X \ \textit{in } \mathcal{L}^q$$

Further, no other implications hold in general.

Proof. Suppose that $X_n \to X$ a.s.. We will show that this implies that $X_n \to X$ in probability. Note, for any $\varepsilon > 0$, we have that

$$\mathbb{P}(|X_n - X| \geq \varepsilon \text{ for infinitely many } n) = \mathbb{P}\Big(\bigcap_m \bigcup_{n \geq m} \{|X_n - X| \geq \varepsilon\}\Big) = 0.$$

Now,

$$\mathbb{P}(|X_m - X| \geq \varepsilon) \leq \mathbb{P}\Big(\bigcup_{n \geq m} \{|X_n - X| \geq \varepsilon\}\Big).$$

Taking the limit $m \to \infty$, we get

$$\begin{aligned}\lim_{m\to\infty} \mathbb{P}(|X_m - X| \geq \varepsilon) &\leq \lim_{m\to\infty} \mathbb{P}\Big(\bigcup_{n \geq m} \{|X_n - X| \geq \varepsilon\}\Big) \\ &= \mathbb{P}\Big(\bigcap_m \bigcup_{n \geq m} \{|X_n - X| \geq \varepsilon\}\Big) = 0,\end{aligned}$$

as required.

An example of a sequence that converges in probability, but not almost surely, will be discussed on Problem Sheet 6. The fact that convergence in $\mathcal{L}^1$ implies convergence in probability will proved on Problem Sheet 7. All other implications and counter examples can be found in [GS01, Chapter 7]. □

In special cases, we can formulate the converse statements of the above as well.

Theorem 8.4.

(i) If $X_n \xrightarrow{\mathcal{D}} c$ where c is a constant, then $X_n \xrightarrow{prob.} c$;

(ii) If $X_n \xrightarrow{prob.} X$ and $\mathbb{P}(|X_n| \leq K) = 1$ $\forall n$ for some constant K, then $X_n \to X$ in $\mathcal{L}^p$ for all $p \geq 1$;

(iii) If

$$\sum_{n=1}^{\infty} \mathbb{P}\big(|X_n - X| > \varepsilon\big) < \infty$$

for all $\varepsilon > 0$, then $X_n \to X$ almost surely.

The following proof is not examinable!

Proof. (i) Note that the distribution function of a random variable X that is constant equal to c is given as

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 0 & \text{if } x < c \\ 1 & \text{if } x \geq c \end{cases}$$

In particular, if $X_n \to c$ in distribution, then as $n \to \infty$,

$$\mathbb{P}(X_n \le x) \to \begin{cases} 0 & \text{if } x < c \\ 1 & \text{if } x \ge c \end{cases}$$

Therefore,

$$\begin{aligned} \mathbb{P}(|X_n - c| > \varepsilon) &= \mathbb{P}(X_n < c - \varepsilon) + \mathbb{P}(X_n > c + \varepsilon) \\ &\le \mathbb{P}(X_n \le c - \varepsilon) + 1 - \mathbb{P}(X_n \ge c + \varepsilon) \to 0 + 1 - 1 = 0\,, \end{aligned}$$

as $n \to \infty$, which shows that $X_n \to c$ in probability.

(ii) Suppose that $X_n \to X$ in probability. Then, since $\mathbb{P}(|X_n| \le K) = 1$, it follows that $\mathbb{P}(|X| \le K) = 1$. Indeed, using that $|X_n| \le |X_n - X| + |X|$,

$$\begin{aligned} \mathbb{P}(|X| \ge K + 2\varepsilon) &\le \mathbb{P}(|X| \ge K + 2\varepsilon; |X_n - X| \le \varepsilon) + \mathbb{P}(|X_n - X| \ge \varepsilon) \\ &= \mathbb{P}(K + 2\varepsilon \le |X| \le |X_n - X| + |X_n| \le \varepsilon + K) + \mathbb{P}(|X_n - X| \ge \varepsilon) \\ &= \mathbb{P}(|X_n - X| \ge \varepsilon). \end{aligned}$$

Hence, as $n \to \infty$, we deduce that $\mathbb{P}(|X| \ge K + 2\varepsilon) = 0$. Then, we can let $\varepsilon \downarrow 0$ to find that $\mathbb{P}(|X| > K) = 0$.

Now, we can argue that on a set of probability 1

$$|X_n - X|^p \le \varepsilon^p \mathbb{1}_{\{|X_n - X| \le \varepsilon\}} + (2K)^p \mathbb{1}_{\{|X_n - X| > \varepsilon\}}.$$

Taking expectations, we obtain

$$\mathbb{E}(|X_n - X|^p) \le \varepsilon^p + (2K)^p \mathbb{P}(|X_n - X| > \varepsilon).$$

Hence, letting $n \to \infty$ gives

$$\lim_{n\to\infty} \mathbb{E}|X_n - X|^p \le \varepsilon^p.$$

Since the left hand side does not depend on ε, we have that $\lim_{n\to\infty} \mathbb{E}|X_n - X|^p = 0$.

(iii) Note that for any $\varepsilon > 0$,

$$\begin{aligned} \mathbb{P}(|X_n - X| > \varepsilon \text{ for infinitely many } n) &= \mathbb{P}\Big(\bigcap_{m\in\mathbb{N}} \bigcup_{n \ge m} |X_n - X| \ge \varepsilon \Big) \\ &= \lim_{m\to\infty} \mathbb{P}\Big(\bigcup_{n\ge m} |X_n - X| \ge \varepsilon \Big) \\ &\le \lim_{m\to\infty} \sum_{n \ge m} \mathbb{P}(|X_n - X| \ge \varepsilon) = 0, \end{aligned}$$

since by assumption $\sum_{n\in\mathbb{N}} \mathbb{P}(|X_n - X| \ge \varepsilon) < \infty$. Now, finally,

$$\begin{aligned} \mathbb{P}(X_n \not\to X) &\le \mathbb{P}\Big(\bigcup_{k\in\mathbb{N}} \{|X_n - X| \ge \frac{1}{k} \text{ infinitely often}\} \Big) \\ &\le \sum_{k\in\mathbb{N}} \mathbb{P}(|X_n - X| \ge \frac{1}{k} \text{ infinitely often}) = 0, \end{aligned}$$

by the above, showing that $X_n \to X$ a.s. □

9 Conditional Expectation

9.1 Discrete Random Variables

Recall that if A is an event with $\mathbb{P}(A) > 0$ and X is a non-negative random variable taking values in the discrete set $X(\Omega) := \{x_1, x_2, \dots\}$, then

$$\mathbb{E}(X|A) = \frac{\mathbb{E}(X\mathbb{1}_A)}{\mathbb{P}(A)}.$$

Note that

$$\begin{aligned}\mathbb{E}(X|A) = \frac{\mathbb{E}(X\mathbb{1}_A)}{\mathbb{P}(A)} &= \sum_{j=1}^{\infty} \frac{\mathbb{E}(X\mathbb{1}_{\{X=x_j\}}\mathbb{1}_A)}{\mathbb{P}(A)} \\ &= \sum_{j=1}^{\infty} x_j \frac{\mathbb{E}(\mathbb{1}_{\{X=x_j\}}\mathbb{1}_A)}{\mathbb{P}(A)} = \sum_{j=1}^{\infty} x_j \mathbb{P}(X = x_j|A).\end{aligned}$$

This is just the mean of the conditional distribution of X given the event A occurs.

Definition 9.1 (Conditional expectation given a random variable)**.** For a discrete random variable Y taking values in $Y(\Omega) := \{y_1, y_2, \dots\}$ where $\mathbb{P}(Y = y_j) > 0$ and $\sum_{j=1}^{\infty} \mathbb{P}(Y = y_j) = 1$, we define the *conditional expectation of X given Y* as

$$\mathbb{E}(X|Y) := \sum_{j=1}^{\infty} \mathbb{1}_{\{Y=y_j\}} \mathbb{E}(X|Y = y_j).$$

Remark 9.2. (i) Note that $\mathbb{E}(X|Y = y_j)$ is just a number, whereas $\mathbb{E}(X \mid Y)$ is a random variable since the indicators are random variables.

(ii) Furthermore, the random variable $\mathbb{E}(X|Y)$ is a function of the random variable Y. Indeed, define $f(y) := \mathbb{E}(X|Y = y)$ where $\mathbb{P}(Y = y) > 0$, then on the event $Y = y_i$, we have that
$$\mathbb{E}(X \mid Y) = \mathbb{E}(X \mid Y = y_i) = f(y_i) = f(Y).$$
Since, $\bigcup_i \{Y = y_i\} = \Omega$, we have that $\mathbb{E}(X \mid Y) = f(Y)$.

Example 9.3. Suppose at a nuclear power plant there are Y radiation alerts, $Y \sim \mathrm{Po}(\lambda)$. Each radiation alert results in a radiation leak with probability p independent of the others and of Y.

(i) Given $Y = n$, what is the expected number of radiation leaks?

(ii) Conditional on the random variable Y, what is the expected number of radiation leaks?

Solution. Let Z be the number of radiation leaks.

(i) Given $Y = n$, $Z \sim B(n,p)$. Hence $\mathbb{E}(Z|Y = n) = np$, which is a deterministic number.

(ii) Then, the conditional expectation of Z given the random variable Y is given by $\mathbb{E}(Z|Y) = pY$, i.e. for $\omega \in \Omega$, $\mathbb{E}(Z|Y)(\omega) := pY(\omega)$. Note that $\mathbb{E}(Z|Y)$ is a random variable only dependent on Y.

Lemma 9.4 (The tower property). *We have that* $\mathbb{E}\big(\mathbb{E}(X|Y)\big) = \mathbb{E}(X)$.

Proof. Suppose Y takes values in $Y(\Omega) := \{y_1, y_2, \dots\}$. Then

$$\begin{aligned}
\mathbb{E}\big(\mathbb{E}(X|Y)\big) &= \mathbb{E}\Big(\sum_{y_j \in Y(\Omega)} \mathbb{1}_{\{Y=y_j\}} \mathbb{E}(X|Y = y_j)\Big) \\
&= \sum_{y_j} \mathbb{E}\left(\mathbb{1}_{\{Y=y_j\}} \mathbb{E}(X|Y = y_j)\right) \\
&= \sum_{y_j} \mathbb{E}(X|Y = y_j) \mathbb{E}\big(\mathbb{1}_{\{Y=y_j\}}\big) \qquad \text{[since } \mathbb{E}(X \mid Y = y_j) \text{ is a constant]} \\
&= \sum_{y_j} \mathbb{E}(X|Y = y_j) \mathbb{P}(Y = y_j) \\
&= \sum_{y_j} \mathbb{E}(X \mathbb{1}_{\{Y=y_j\}}) \\
&= \mathbb{E}\Big(X \sum_{y_j} \mathbb{1}_{\{Y=y_j\}}\Big) \qquad \text{[by definition of the conditional expectation]} \\
&= \mathbb{E}(X).
\end{aligned}$$

□

Example 9.5. What is the distribution of the number of radiation leaks from example 9.3?

Solution. Let $X_1, X_2, \dots$ be i.i.d. Ber(p). Then

$$Z = \sum_{j=1}^{Y} X_j$$

Now for $q := 1 - p$,

$$\mathbb{E}(s^Z|Y = n) = (ps + q)^n$$

the PGF of a $B(n,p)$. Then $\mathbb{E}(s^Z|Y) = (ps + q)^Y$. This is a random variable depending on Y. Now, using the tower property and the fact that $\mathbb{E}(t^Y) = e^{\lambda(t-1)}$ for the Poisson random variable Y,

$$\begin{aligned}\mathbb{E}(s^Z) &= \mathbb{E}\big(\mathbb{E}(s^Z|Y)\big) = \mathbb{E}\big((ps+q)^Y\big)\\ &= \exp\{\lambda((ps+q)-1)\} = \exp\{\lambda p(s-1)\}\end{aligned}$$

but this is the PGF of a $\mathrm{Po}(\lambda p)$ distribution, hence $Z \sim \mathrm{Po}(\lambda p)$.

Lemma 9.6 (Other key properties)**.** *For discrete random variables X, Y, Z and a Borel function h ,*

(i) 'Taking out what is known': $\mathbb{E}\big(h(Y)X|Y\big) = h(Y)\mathbb{E}(X|Y)$.

(ii) If X and Y are independent, then $\mathbb{E}(X|Y) = \mathbb{E}(X)$.

(iii) Linearity: For any constants $\lambda, \mu \in \mathbb{R}$, $\mathbb{E}(\lambda X + \mu Z|Y) = \lambda\mathbb{E}(X|Y) + \mu\mathbb{E}(Z|Y)$.

Proof. Recall, that for discrete random variables, we have that

$$\mathbb{E}(X|Y) := \sum_{y_j \in Y(\Omega)} \mathbb{1}_{\{Y=y_j\}}\mathbb{E}(X|Y=y_j)$$

(i) We have the following, using the definitions of the conditional expectations,

$$\begin{aligned}\mathbb{E}\big(h(Y)X|Y\big) &= \sum_j \mathbb{1}_{\{Y=y_j\}}\mathbb{E}\big(h(Y)X|Y=y_j\big)\\ &= \sum_j \mathbb{1}_{\{Y=y_j\}}\frac{\mathbb{E}\big(h(Y)X\mathbb{1}_{\{Y=y_j\}}\big)}{\mathbb{P}(Y=y_j)}\\ &= \sum_j \mathbb{1}_{\{Y=y_j\}}\frac{\mathbb{E}\big(h(y_j)X\mathbb{1}_{\{Y=y_j\}}\big)}{\mathbb{P}(Y=y_j)}\\ &= \sum_j \mathbb{1}_{\{Y=y_j\}}h(y_j)\frac{\mathbb{E}(X\mathbb{1}_{\{Y=y_j\}})}{\mathbb{P}(Y=y_j)}\\ &= \sum_j \mathbb{1}_{\{Y=y_j\}}h(Y)\mathbb{E}(X|Y=y_j)\\ &= h(Y)\sum_j \mathbb{1}_{\{Y=y_j\}}\mathbb{E}(X|Y=y_j)\\ &= h(Y)\mathbb{E}(X|Y).\end{aligned}$$

(ii) Since X and Y are independent, we have that for $X(\Omega) = \{x_1, x_2, \ldots\}$,

$$\mathbb{E}(X|Y=y_j) = \sum_i x_i\,\mathbb{P}(X=x_i|Y=y_j) = \sum x_i\,\mathbb{P}(X=x_i) = \mathbb{E}(X)$$

Hence

$$\mathbb{E}(X|Y) = \sum_j \mathbb{1}_{\{Y=y_j\}}\mathbb{E}(X|Y=y_j) = \sum_j \mathbb{1}_{\{Y=y_j\}}\mathbb{E}(X)$$
$$= \mathbb{E}(X)\sum_j \mathbb{1}_{\{Y=y_j\}} = \mathbb{E}(X)$$

(iii) Follows directly from the linearity of $\mathbb{E}(\cdot\,|\,Y=y_j)$. □

9.2 Continuous Random Variables

Suppose X and Y have a joint PDF $f_{X,Y}(x,y)$ on $\mathbb{R}^2$, then, for any $A \in \mathcal{B}(\mathbb{R}^2)$:

$$\mathbb{P}((X,Y) \in A) = \int_{(x,y)\in A} f_{X,Y}(x,y)\,dy\,dx$$

Recall that we can recover the *marginal distribution* of one of the variables, e.g. X, by integrating out the other variables, i.e. for any $A \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}(X \in A) = \int_{y\in\mathbb{R},x\in A} f_{X,Y}(x,y)\,dx\,dy = \int_{x\in A}\int_{y\in\mathbb{R}} f_{X,Y}(x,y)\,dy\,dx = \int_{x\in A} f_X(x)\,dx,$$

if we define

$$f_X(x) := \int_{y\in\mathbb{R}} f_{X,Y}(x,y)\,dy,$$

the marginal density of X. The definition of f_Y follows analogously as

$$f_Y(y) := \int_{x\in\mathbb{R}} f_{X,Y}(x,y)\,dx.$$

Definition 9.7. We define the conditional PDF $f_{X|Y}(x|y)$ of X given that $Y=y$ via

$$f_{X|Y}(x|y) := \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

for any y such that $f_Y(y) > 0$ and set it to be 0 otherwise. The conditional distribution of X given $Y=y$ is given by

$$F_{X|Y}(x|y) = \int_{-\infty}^{x} f_{X|Y}(u|y)\,\mathrm{d}u =: \mathbb{P}(X \le x|Y=y)$$

Now let

$$\Psi(y) := \mathbb{E}(X|Y=y) := \int_{x\in\mathbb{R}} x\, f_{X|Y}(x|y)\,\mathrm{d}x,$$

then the conditional expectation $\mathbb{E}(X|Y)$ of X given Y is given by $\Psi(Y)$.

As in the discrete case, $\mathbb{E}(X|Y)$ is a random variable and a function of the random variable Y only.

Note that the same properties as listed in Lemmas 9.4 and 9.6 hold as in the discrete case. For example, we have with Ψ as above:

$$\begin{aligned}
\mathbb{E}(\,\mathbb{E}(X\,|\,Y)) = \mathbb{E}(\Psi(Y)) &= \int_{\mathbb{R}} \Psi(y)\, f_Y(y)\, dy \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}} x f_{X|Y}(x|y)\, dx f_Y(y)\, dy \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}} x \frac{f_{X,Y}(x,y)}{f_Y(y)}\, dx \mathbb{1}_{\{f_Y(y)>0\}} f_Y(y)\, dy \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}} x f_{X,Y}(x,y)\, dx \mathbb{1}_{\{f_Y(y)>0\}}\, dy \\
&= \int_{\mathbb{R}} x \int_{\mathbb{R}} f_{X,Y}(x,y)\, dy\, dx \\
&= \int_{\mathbb{R}} x f_X(x)\, dx = \mathbb{E}(X).
\end{aligned}$$

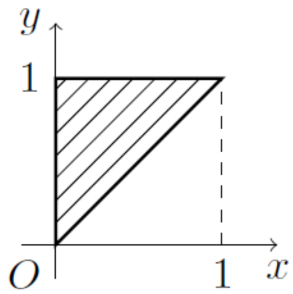
Here, we used that for any y.

$$f_Y(y) = 0 \implies \int_{\mathbb{R}} f_{X,Y}(x,y)\, dx = 0 \implies \int_{\mathbb{R}} x f_{X,Y}(x,y) = 0.$$

Example 9.8. Let X and Y have joint PDF given by

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{y}, & 0 \le x \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

The region, where $f_{X,Y}(x,y) > 0$, can be seen in the following diagram:



So, the marginal PDF of Y, $f_Y(y)$, is given by

$$f_Y(y) = \int_{x\in\mathbb{R}} f_{X,Y}(x,y)\, \mathrm{d}x = \int_0^y \frac{1}{y} \mathrm{d}x = \frac{1}{y} \int_0^y 1 \mathrm{d}x = 1$$

for $y \in [0,1]$. That is, $Y \sim \text{Unif}[0,1]$. Then, the conditional PDF $f_{X|Y}(x|y)$, of X given $Y = y \in [0,1]$ is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \begin{cases} \frac{1}{y}, & 0 \le x \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

That is, given $Y = y \in [0,1]$, we have $X \sim \text{Unif}[0,y]$. In particular, $\mathbb{E}(X|Y = y) = y/2$ for $y \in [0,1]$, and hence $\mathbb{E}(X|Y) = Y/2$ - a random variable whose randomness depends only on the random variable Y.

10 Branching Processes

10.1 The Model

Consider a model for the evolution of bacteria. The colony evolves in discrete time. Each bacterium lives for 1 unit of time, at the end of which it is replaced by a random number of descendants. For different bacteria, the number of replacements are independent and have the same distribution as a random variable X, where

$$\mathbb{P}(X = k) = p_k, \qquad \text{for } k = 0, 1, 2, \dots.$$

The distribution of X is called the *offspring distribution* and p_k is the probability that one bacterium is being replaced by k bacteria.

More formally, let $(X_i^{(n)}, i \in \mathbb{N}, n \in \mathbb{N})$ be i.i.d. random variables with the same distribution as X. These random variables have the following interpretation: the ith bacterium (if it exists) in generation $n-1$ will be replaced by $X_i^{(n)}$ bacteria in generation n. Let Z_n represent the number of bacteria in generation n. Under the probability law $\mathbb{P}$, and expectation $\mathbb{E}$, we suppose that $Z_0 = 1$. Then,

$$Z_{n+1} = X_1^{(n+1)} + X_2^{(n+1)} + \cdots + X_{Z_n}^{(n+1)} = \sum_{i=1}^{Z_n} X_i^{(n+1)}.$$

Morever, we assume throughout that

$$\mu := \mathbb{E}(X) = \sum_{k=0}^{\infty} k p_k < \infty$$

In the following the two main questions that we will answer are:

1. What is the average size of the population?
2. What is the probability the population dies out?

10.2 Average Size of the Population

Our first goal is to find the average size of the population. We start by finding the conditional size of the next generation given we know the size of the current one. For each fixed $n, r \in \mathbb{Z}^+$,

$$\begin{aligned}
\mathbb{E}(Z_{n+1} \,|\, Z_n = r) &= \mathbb{E}\left(\sum_{i=1}^{Z_n} X_i^{(n+1)} \,\middle|\, Z_n = r \right) \\
&= \mathbb{E}\left(\sum_{i=1}^{r} X_i^{(n+1)} \,\middle|\, Z_n = r \right) \\
&= \mathbb{E}\left(\sum_{i=1}^{r} X_i^{(n+1)} \right) \qquad \text{[by independence]} \\
&= \sum_{i=1}^{r} \mathbb{E}\left(X_i^{(n+1)} \right) \\
&= \sum_{i=1}^{r} \mathbb{E}(X) = \mu r
\end{aligned}$$

since $\mu := \mathbb{E}(X)$ and since $X_1^{(n+1)}, X_2^{(n+1)}, \ldots$ are i.i.d. with the same distribution as X. Then, if define

$$f(r) := \mathbb{E}(Z_{n+1} \,|\, Z_n = r) = \mu r.$$

Then, we know that

$$\mathbb{E}(Z_{n+1} \,|\, Z_n) = f(Z_n) = \mu Z_n.$$

Hence, using the tower property, Lemma 9.4, we can deduce that

$$\begin{aligned}
\mathbb{E}(Z_{n+1}) &= \mathbb{E}\big(\mathbb{E}(Z_{n+1} \,|\, Z_n)\big) \\
&= \mathbb{E}(\mu Z_n) = \mu \mathbb{E}(Z_n).
\end{aligned}$$

Hence,

$$\mathbb{E}(Z_{n+1}) = \mu \mathbb{E}(Z_n) = \mu^2 \mathbb{E}(Z_{n-1}) = \cdots = \mu^{n+1} \mathbb{E}(Z_0) = \mu^{n+1}$$

since $\mathbb{E}(Z_0) = 1$. In particular, we have shown the following:

Proposition 10.1. *For any $n \geq 0$,*

$$\mathbb{E}(Z_n) = \mu^n \quad \forall n \geq 0$$

Note that if $\mu > 1$, then $\mathbb{E}(Z_n) \to +\infty$ and if $\mu < 1$, then $\mathbb{E}(Z_n) \to 0$. If $\mu = 1$, then $\mathbb{E}(Z_n) = 1$ for all $n \in \mathbb{N}$.

10.3 Extinction probability of the population

We want to find

$$\pi := \mathbb{P}(\text{ultimate extinction}) = \mathbb{P}(Z_n = 0 \text{ for some n})$$

Assume that $\mathbb{P}(X = 0) =: p_0 > 0$ throughout (otherwise trivially, $\pi = 0$). Since $\mathbb{E}(Z_n) = \mu^n$, it is plausible that $\pi = 1$ if $\mu < 1$ and $\pi < 1$ if $\mu > 1$. Define

$$\pi_n := \mathbb{P}(\text{extinction by time } n) = \mathbb{P}(Z_n = 0)$$

Lemma 10.2. *We have that* $\pi_n \uparrow \pi$ *and*

$$\pi_{n+1} = g(\pi_n),$$

where

$$g(\theta) = \mathbb{E}(\theta^X),$$

is the PGF of the offspring distribution.

Proof. Note that $\{Z_{n+1} = 0\} \supseteq \{Z_n = 0\}$ since if bacteria are extinct at time n, they are also extinct at time $n+1$. Then,

$$\{Z_n = 0\} \uparrow \bigcup_{i=0}^{\infty} \{Z_i = 0\} = \{Z_k = 0 \text{ for some } k\}$$

and continuity of $\mathbb{P}$, or MON, tell us that

$$\pi_n = \mathbb{P}(Z_n = 0) \uparrow \mathbb{P}(Z_n = 0 \text{ for some } n) = \pi,$$

which proves the first part of the lemma. Now, by splitting over the number of bacteria alive in the first generation, we get

$$\begin{aligned}
\pi_{n+1} &= \mathbb{P}(Z_{n+1} = 0) \\
&= \sum_{k=0}^{\infty} \mathbb{P}(Z_1 = k; Z_{n+1} = 0) \\
&= \sum_{k=0}^{\infty} \mathbb{P}(Z_1 = k)\mathbb{P}(Z_{n+1} = 0 \,|\, Z_1 = k).
\end{aligned}$$

But $\mathbb{P}(Z_{n+1} = 0 \,|\, Z_1 = k)$ is the probability that k independent family trees each starting from one bacterium all die out before generation n, so this is π_n^k. Hence

$$\pi_{n+1} = \sum_{k=0}^{\infty} \mathbb{P}(X = k)\pi_n^k = g(\pi_n),$$

proving the second part of the lemma. $\square$

Taking the limit $n \to \infty$ in the previous lemma, suggests that π is a fixed point of g, i.e. $g(\pi) = \pi$. Indeed, this fact can be used to characterize the extinction probability:

Theorem 10.3. *If $\pi := \mathbb{P}(extinction)$, then*

(i) if $\mu := \mathbb{E}(X) \leq 1$, we have $\pi = 1$;

(ii) if $\mu > 1$, then π is the unique root of $g(\pi) = \pi$ in $(0, 1)$.

Proof. By Lemma 10.2 and since $g(\theta)$ is continuous,

$$\lim_{n\to\infty} \pi_{n+1} = \lim_{n\to\infty} g(\pi_n) = g\left(\lim_{n\to\infty} \pi_n\right)$$

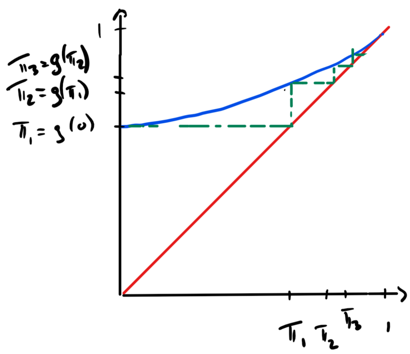
Therefore, $\pi = g(\pi)$.
Also note that $g'(1) = \mathbb{E}(X) =: \mu$ and

$$g(1) = \sum_k \mathbb{P}(X = k) = 1, \quad g(0) = p_0 > 0.$$

Furthermore,

$$g'(\theta) = \mathbb{E}(X\theta^{X-1}) > 0, \quad \text{and} \quad g''(\theta) = \mathbb{E}\left(X(X-1)\theta^{X-2}\right) \geq 0.$$

In particular, $g(\theta)$ is convex; its slope, $g'(\theta)$, is never decreasing. There are two situations that can arise:
Case 1: $g'(1) = \mu \leq 1$



We first claim that $g'(\theta) < 1$ for all $\theta \in [0, 1)$. If $\mu < 1$, then this follows immediately: using that $g'(\theta)$ is increasing, we have $g'(\theta) \leq g'(1) = \mu < 1$ for all $\theta \in [0, 1)$.
If $\mu = 1$, then

$$1 = g'(1) = p_1 + 2p_2 + 3p_3 + \cdots \tag{20}$$

implies that $p_1 + p_0 < 1$. Indeed, if $p_1 + p_0 = 1$, then $p_2 = p_3 = \ldots = 0$. Therefore, (20) implies that $p_1 = 1$. However, since $p_0 > 0$, we know that $p_1 < 1$, a contradiction.

Now, $p_1 + p_0 < 1$ implies that there exists $\ell \geq 2$ such that $p_\ell > 0$ and therefore

$$g''(\theta) = \sum_{k \geq 2} k(k-1) p_k \theta^{k-2} > 0,$$

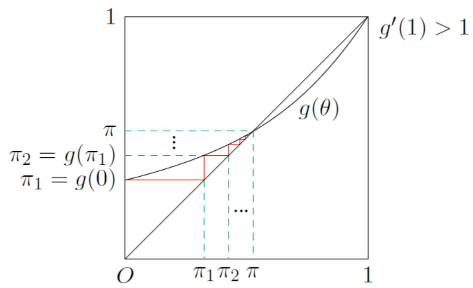
and thus g' is strictly increasing so that $g'(\theta) < g'(1) = 1$ for any $\theta \in [0, 1)$.

Now, assume that there exists $\theta' \in [0, 1)$ such that $g(\theta') = \theta'$. Then, by the mean value theorem, there exists $\theta^* \in (\theta', 1)$ such that

$$g'(\theta^*) = \frac{g(1) - g(\theta')}{1 - \theta'} = \frac{1 - \theta'}{1 - \theta'} = 1. \tag{21}$$

This contradicts our initial claim and so $g(\theta) > \theta$ for all $\theta \in [0, 1)$. Hence, we must have $\pi = g(\pi)$ implies that $\pi = 1$.

Case 2: $g'(1) = \mu > 1$:



Note that in this case there has to be at least one $\theta \in [0, 1)$ such that $g(\theta) = \theta$. Otherwise, $g(0) > 0$ would imply $g(\theta) \geq \theta$ for all $\theta \in [0, 1]$, which would mean that $g'(1) \leq 1$. Also, there cannot be more than one such point. Indeed, $\mu \geq 1$ means that there exists $\ell \geq 2$ such $p_\ell > 0$ so that $g''(\theta) > 0$ and g' is strictly increasing as above. But as in (21) between any two fixed points of g (i.e. any θ such that $g(\theta) = \theta$) there is a point θ^* with $g'(\theta^*) = 1$. Thus, g' strictly increasing implies that there are at most two fixed points (one of which is 1).

Let $\pi_\infty \in (0, 1)$ be this point with $g(\pi_\infty) = \pi_\infty$. We want to show that

$$\pi := \mathbb{P}(\text{extinction}) = \pi_\infty \in (0, 1)$$

From the picture, it is clear that $\pi_n \uparrow \pi_\infty$, but we already know that

$$\pi_n \uparrow \pi = \mathbb{P}(\text{extinction})$$

so we should have $\pi = \pi_\infty \in (0, 1)$, the unique root in $(0, 1)$ of $\pi = g(\pi)$.

To argue more carefully,

$$0 < \pi_1 = \mathbb{P}(Z_1 = 0) = p_0 = g(0) \leq g(\pi_\infty) = \pi_\infty$$

as g is increasing on $[0,1]$ and $\pi_\infty \in (0,1)$ so $\pi_\infty > 0$. That is, $\pi_1 \leq \pi_\infty$. Again, since g is increasing,

$$\begin{aligned} \pi_n \leq \pi_\infty &\implies g(\pi_n) \leq g(\pi_\infty) \\ &\implies \pi_{n+1} \leq \pi_\infty \end{aligned}$$

since $\pi_{n+1} = g(\pi_n)$ and $\pi_\infty = g(\pi_\infty)$. By induction, $\pi_n \leq \pi_\infty \; \forall n \geq 1$, so

$$\pi = \lim_{n\to\infty} \pi_n \leq \pi_\infty < 1$$

by definition of π_∞. But $\pi = g(\pi)$ and we now know that $\pi < 1$ and thus it follows that $\pi = \pi_\infty$. $\square$

11 Poisson Processes

A Poisson process on $[0,\infty)$ can be used to model random occurrences of events such as the times of arrivals at a queue, call requests at a telephone exchange, times of clicks of a Geiger counter.

In the most basic case, the time parameter set for arrivals is $[0,\infty)$. Let $N(A)$ denote the number of arrivals in a time interval $A \subset [0,\infty)$, e.g. we write $N(t, t+s]$ for the number of arrivals during time $(t, t+s]$.

11.1 Poisson Point Processes on $[0,\infty)$

The first description of a Poisson is as follows:

Definition 11.1. A *Poisson process* of rate λ, denoted $\mathrm{PP}(\lambda)$, is a collection of random variables N_I with $I = (s,t]$, for $0 \le s \le t$, where N_I counts the number of arrival during time interval I, that satisfies:

(i) the number of arrivals $N(A_1), N(A_2), \dots, N(A_n)$ during disjoint time intervals $A_1, A_2, \dots, A_n$ are independent random variables;

(ii) the number of arrivals during any interval of length t has a $\mathrm{Po}(\lambda t)$ distribution, that is, $N(A_i) \sim \mathrm{Po}(\lambda|A_i|)$ where $|A_i|$ is the length of the time interval A_i.

Note that if $X \sim \mathrm{Po}(\lambda)$ and $Y \sim \mathrm{Po}(\mu)$ are independent random variables, we know that $X+Y \sim \mathrm{Po}(\lambda+\mu)$. This is consistent with our definition as $N(0,t] + N(t,t+s] = N(0,t+s]$.

An equivalent description of a Poisson process is the following *inter-arrival description*:

Lemma 11.2. *A Poisson process of rate λ, $\mathrm{PP}(\lambda)$, is fully described by the fact that the inter-arrival times $\xi_1, \xi_2, \dots$ are i.i.d. exponential random variables with parameter λ. That is, arrivals occur at times $T_1 := \xi_1, T_2 := \xi_1 + \xi_2, \dots, T_n := \sum_{i=1}^n \xi_i$, and then the number of arrivals by time t is given by*

$$N(0,t] := \max\left\{ n : \sum_{k=1}^{n} \xi_k \le t \right\}$$

Moreover, $N(s,t]$ can then be recovered as $N(s,t] = N(0,t] - N(0,s]$.

This construction is illustrated in Figure 5.

Proof. We will not prove this result. However, we will point out a few connections.

First of all assume that $N_{(s,t]}$ is a Poisson process and consider the first arrival time

$$\xi_1 := \inf\{t \ge 0 \,:\, N(0,t] = 1\}.$$

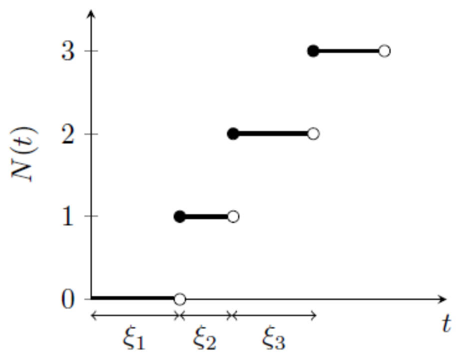


Figure 5: A realisation of a Poisson point process.

Then, since $N(0,t] \sim \text{Po}(\lambda t)$,

$$\mathbb{P}(\xi_1 > t) = \mathbb{P}(N(0,t] = 0) = e^{-\lambda t}.$$

Hence, ξ_1 is $\text{Exp}(\lambda)$-distributed. The converse direction uses that the distribution of T_n is explicitly known. Furthermore, the memory-less property of the exponential distribution is essential for showing the independence of increments. $\square$

11.2 Decomposing a Poisson Process

In the following we will consider a Poisson Process that models arrivals of two different types. In order to obtain an elegant description of this process, we will need the following more basic lemma.

Lemma 11.3 ('Boys and Girls' principle.)**.** *The follow two models are equivalent:*

(i) The total number of births is $\text{Po}(\gamma)$ *and each birth, independently of everything else, produces a boy with probability* p*, or a girl with probability* $q := 1 - p$*.*

(ii) The number of boys is $\text{Po}(\gamma p)$ *and the number of girls is* $\text{Po}(\gamma q)$*, and the number of boys is independent from the number of girls.*

Proof. [(ii) $\implies$ (i)]: If $B \sim \text{Po}(\gamma p)$ and $G \sim \text{Po}(\gamma q)$ and B, G are independent, then we have that $B+G \sim \text{Po}(\gamma p+\gamma q) = \text{Po}(\gamma)$ by Example 5.6. Now,

$$\begin{aligned}
\mathbb{P}(B=b \,|\, B+G=n) &= \frac{\mathbb{P}(B=b; G=n-b)}{\mathbb{P}(B+G=n)} \\
&= \frac{\mathbb{P}(B=b)\mathbb{P}(G=n-b)}{\mathbb{P}(B+G=n)} \\
&= \frac{(\gamma p)^b}{b!}e^{-\gamma p}\frac{(\gamma q)^{n-b}}{(n-b)!}e^{-\gamma q}\frac{n!}{\gamma^n e^{-\gamma}} \\
&= \binom{n}{b}p^b q^{n-b}.
\end{aligned}$$

Thus, $B \sim B(n,p)$ conditional on $B+G=n$, that is, it has the same distribution as if each birth is a boy with probability p and the trials are independent.

[(i) $\implies$ (ii)]: Denote by B, resp. G, the number of boys, resp. girls, born and set $N = B+G$. Using that N is Poisson(γ) and each birth is independently a boy with probability p

$$\begin{aligned}
\mathbb{P}(B=b; G=g) &= \mathbb{P}(N=b+g; B=b) \\
&= \frac{\gamma^{b+g}}{(b+g)!}e^{-\gamma}\binom{b+g}{b}p^b q^g \\
&= \frac{(\gamma p)^b}{b!}e^{-p\gamma}\frac{(\gamma q)^g}{g!}e^{-q\gamma} \\
&= \mathbb{P}(N_p=b)\mathbb{P}(N_q=g),
\end{aligned}$$

where N_p and N_q are independent random variables with distributions $\text{Po}(\gamma p)$ and $\text{Po}(\gamma q)$ respectively.

Summing over all possible values of g, we find

$$\mathbb{P}(B=b) = \sum_{g\in\mathbb{Z}_+}\mathbb{P}(B=b; G=g) = \mathbb{P}(N_p=b).$$

Showing that $B \sim \text{Po}(\lambda p)$. Similarly, we see that $G \sim \text{Po}(\lambda q)$. Moreover, the above shows that G and B are independent. □

The previous lemma motivates the 'Boys and Girls' principle for Poisson processes.

Proposition 11.4. *The following two models are equal in distribution:*

(Model 1) Men arrive at a queue as a $\text{PP}(\lambda)$ *process, and women arrive at the queue as a* $\text{PP}(\mu)$ *process and these two Poisson processes are independent.*

(Model 2) People arrive at the queue as a $\text{PP}(\lambda+\mu)$ *process and each person, independently of everything else, is a man with probability* $\frac{\lambda}{\lambda+\mu}$ *and a woman with probability* $\frac{\mu}{\lambda+\mu}$.

11.3 Conditional Uniformity

Theorem 11.5. *Let N be a Poisson process of constant rate $\lambda > 0$ on $[0, \infty)$. Conditional on $N(0,t] = n$, the times of the n arrivals over $(0,t]$ are independent uniformly distributed on $(0,t]$.*

Proof. Partition $I := (0,t]$ into disjoint intervals $J_1, \dots, J_m$ of lengths $\ell_1, \dots, \ell_m$, i.e.

$$|J_i| = \ell_i \quad \text{and} \quad \sum_{i=1}^{m} \ell_i = t$$

Let $p_i = \ell_i/t$. Suppose that $Z_1, \dots, Z_n$ are i.i.d. $\mathrm{Unif}(0,t]$ random variables. Then, for $k_1, \dots, k_m \in \{0, \dots, n\}$ with $k_1 + \cdots + k_m = n$,

$$\begin{aligned}
\mathbb{P}\left(\bigcap_{r=1}^{m}\{k_r \text{ of the } Z_1, \dots, Z_n \text{ fall in } J_r\}\right) &= \left(\frac{n!}{k_1!k_2!\cdots k_m!}\right) p_1^{k_1} p_2^{k_2} \cdots p_m^{k_m} \\
&= n! \prod_{i=1}^{m} \frac{p_r^{k_r}}{k_r!} \qquad (22)
\end{aligned}$$

where

$$\left(\frac{n!}{k_1!k_2!\cdots k_m!}\right)$$

is the number of ways n objects can be divided into groups of size $k_1, k_2, \dots, k_m$ where $k_1 + \cdots + k_m = n$ and

$$p_1^{k_1} p_2^{k_2} \cdots p_m^{k_m}$$

is the probability of k_1 given the objects all fall in J_1, and k_2 given the objects fall in $J_2, \dots,$ and k_m given the objects fall in J_m.

Let $N(J_r)$ be the number of points of N in J_r, then by the independence (since the J_r are disjoint) we have

$$\begin{aligned}
\mathbb{P}\Big(\bigcap_{r=1}^{m}\{N(J_r) = k_r\}\Big) &= \prod_{r=1}^{m} \mathbb{P}\big(N(J_r) = k_r\big) \\
&= \prod_{r=1}^{m} \frac{(\lambda \ell_r)^{k_r}}{k_r!} e^{-\lambda \ell_r} \\
&= \left(\prod_{r=1}^{m} \frac{(\lambda p_r t)^{k_r}}{k_r!}\right) \exp\left(-\lambda \sum_{r=1}^{m} \ell_r\right) \\
&= \left(\prod_{r=1}^{m} \frac{p_r^{k_r}}{k_r!}\right) e^{-\lambda t} (\lambda t)^{k_1 + \cdots + k_m} && \text{since } \sum_{r=1}^{m} \ell_r = t \\
&= n! \left(\prod_{r=1}^{m} \frac{p_r^{k_r}}{k_r!}\right) \frac{(\lambda t)^n}{n!} e^{-\lambda t} && \text{since } k_1 + \cdots + k_m = n
\end{aligned}$$

where $(*)$ follows by independence as the J_r are disjoint intervals. So for $J_1, \ldots, J_m$ a partition of $(0, t]$ and $\sum_{r=1}^m k_r = n$, we have that

$$\mathbb{P}\left(\bigcap_{r=1}^{m} \{N(J_r) = k_r\} \,\middle|\, N(0,t] = n\right) = \frac{1}{\mathbb{P}(N(0,t] = n)} \mathbb{P}\left(\bigcap_{r=1}^{m} \{N(J_r) = k_r\}; N(0,t] = n\right)$$
$$= n! \prod_{r=1}^{m} \frac{p_r^{k_r}}{k_r!},$$

which agrees with (22), i.e. the probability that the k_r uniforms follow into the rth Interval, for $r = 1, \ldots, m$. □

11.4 Poisson point processes on $\mathbb{R}^n$

We can generalize the definition of a Poisson process on $[0, \infty)$ to arbitrary subsets of $\mathbb{R}^n$.

Definition 11.6. Let $G \subseteq \mathbb{R}^n$ and let $\lambda(\cdot)$ be a non-negative function on G, $\lambda : G \to \mathbb{R}^+$. A Poisson point process on G with intensity function $\lambda(\cdot)$, written as PPP$(\lambda(\cdot))$, is a random collection of points in G such that

(a) if $G_1, G_2, \ldots$ are disjoint subsets of G, then the number $N(G_1), N(G_2), \ldots$ of points occurring in $G_1, G_2, \ldots$ are independent random variables;

(b) If $G_1 \subseteq G$, then $N(G_1)$ has the Poisson distribution of parameter $\int_{G_1} \lambda(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x}$, that is,

$$N(G_k) \sim \mathrm{Po}\Big(\int \cdots \int_{G_k \subset \mathbb{R}^n} \lambda(x_1, \ldots, x_n)\,\mathrm{d}x_1 \ldots \mathrm{d}x_n\Big)$$

Remark 11.7. (a) If we take $n = 1$, $G = [0, \infty) \subset \mathbb{R}$ and further assume that $\lambda(x)$ is constant equal to $\lambda > 0$, then we recover the classical definition of a Poisson process with intensity λ.

(b) Note that if $G_1 \subseteq G$, then $N(G_1) \sim \mathrm{Po}(\int_{G_1} \lambda(\boldsymbol{x})\,d\boldsymbol{x})$ and in particular, we have

$$\mathbb{E}(N(G_1)) = \int_{G_1} \lambda(\boldsymbol{x})\,d\boldsymbol{x}.$$

Therefore, the function λ controls how dense the number of points is in any region of G.

Example 11.8. Consider a PPP on $\mathbb{R}$ with intensity function $\lambda(x) = e^{-\mu|x|}$. Then,

$$\int_{\mathbb{R}} \lambda(x)\,dx = \int_{\mathbb{R}} e^{-\mu|x|}\,dx = 2\int_0^\infty e^{-\mu x}\,dx = \frac{2}{\mu}.$$

In particular, the total number of points in $\mathbb{R}$ satisfies $N(\mathbb{R}) \sim \mathrm{Po}(\int_{\mathbb{R}} \lambda(x)\,dx) = \mathrm{Po}(2/\mu)$. Note that here the total number of points is finite (almost surely).

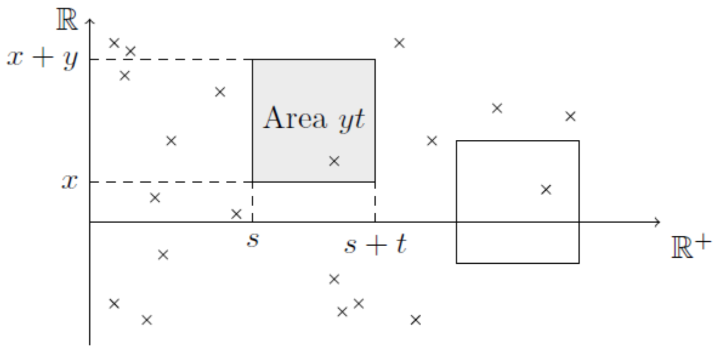
Proposition 11.9 (A conditional uniformity result.)**.** *Suppose we have a* PPP(λ) *on* $G \subseteq \mathbb{R}^n$ *with constant intensity* $\lambda > 0$*. Let* $V \subseteq G$*, then conditional on* $N(V) = n$*, the distribution of these* n *points in* V *is the same as if they were each independently chosen with the uniform distribution on* V*.*

Note that this is really another version of the 'boys and girls' principle.

Example 11.10. We imagine that raindrops land at positions on $\mathbb{R}$ and in time on $[0, \infty)$ as a Poisson Point Process N of constant rate λ on $\mathbb{R}^+ \times \mathbb{R}$ time space. Then the number of raindrops landing in $[x, x+y]$ between times s and $s+t$ is

$$N\big((s, s+t] \times [x, x+y]\big) \sim \text{Po}\left(\int_s^{s+t} \int_x^{x+y} \lambda \, \mathrm{d}x \, \mathrm{d}t\right) = \text{Po}(\lambda y t)$$

This is illustrated in the following diagram:



(a) What's the distribution of the first time a raindrop falls in $[0, x]$? We have that

$$\mathbb{P}(\text{no raindrops land in } [0,x] \text{ by time } t) = \mathbb{P}\big(N((0,t] \times [0,x]) = 0\big) = e^{-\lambda x t}$$

So the time of the first raindrop falling in $[0, x]$ is

$$T_{[0,x]} \sim \text{Exp}(\lambda x)$$

Note that the first drop to arrive anywhere on $[0, \infty)$ does so immediately in time. Also,

$$\mathbb{P}(k \text{ raindrops fall in } [0,x] \text{ by time } t) = \mathbb{P}\big(N((0,t] \times [0,x]) = k\big) = \frac{(\lambda x t)^k}{k!} e^{-\lambda x t}$$

(b) Given that n raindrops have landed in $[0, x]$ by time t, what's the distribution of the first time, U, a raindrop arrives in $[0, x]$?

As arrivals happen at constant intensity λ in $(0, t] \times [0, x]$, then n raindrops are each independent and uniformly distributed in this rectangle, i.e. $(T_i, X_i) \sim \text{Unif}\big((0, t], [0, x]\big)$

for $i = 1, \ldots, n$. Then

$$\begin{aligned}
&\mathbb{P}(U > s \mid N((0,t] \times [0,x]) = n) \\
&\qquad = \mathbb{P}(\text{no points occurred by time } s \mid N((0,t] \times [0,x]) = n) \\
&\qquad = \mathbb{P}(T_i \in [s,t] \text{ for all } i \in \{1, \ldots, n\}) \\
&\qquad = \left(\frac{t-s}{t}\right)^n
\end{aligned}$$

It follows that the conditional distribution function of U is given as

$$\mathbb{P}(U \leq s \mid N((0,t] \times [0,x]) = n) = 1 - \mathbb{P}(U > s \mid N((0,t] \times [0,x]) = n) = 1 - \left(\frac{t-s}{t}\right)^n.$$

Hence, by differentiating with respect to s, we obtain that U has density

$$n\left(\frac{t-s}{t}\right)^{n-1}, \quad s \in [0,t]$$

(c) What's the distribution of the closest raindrop to the origin by time t?

$$\mathbb{P}(\text{no raindrops land in } [-x,x] \text{ by time } t) = \mathbb{P}\big(N((0,t] \times [-x,x]) = 0\big) = e^{-2\lambda x t}$$

so if D_t is the distance to the nearest raindrop by time t, then $\mathbb{P}(D_t > x) = e^{-2\lambda x t}$, i.e. $D_t \sim \mathrm{Exp}(2\lambda t)$.

12 Generating functions and the central limit theorem

We have seen that for non-negative random variables taking values in $\mathbb{Z}_+$ probability generating functions have been very useful. Recall that for a $\mathbb{Z}^+$-valued random variable X, the probability generating function, PGF, is given by

$$g_X(\theta) := \mathbb{E}(\theta^X), \quad 0 \le \theta \le 1$$

There are further generating functions that work for more general random variables.

Definition 12.1. (i) The *moment generating function*, MGF, of a random variable X is given by

$$M_X(\alpha) := \mathbb{E}(e^{\alpha X}),$$

for all $\alpha \in \mathbb{R}$ such that the expectation is finite.

(ii) The *characteristic function*, CF, of a random variable X is given by

$$\varphi_X(\alpha) := \mathbb{E}(e^{i\alpha X}), \quad \text{for all } \alpha \in \mathbb{R}.$$

where $i \in \mathbb{C}$, $i^2 = -1$.

Remark 12.2. Unlike moment generating functions, characteristic functions are always well defined via

$$\mathbb{E}(e^{i\alpha X}) = \mathbb{E}(\cos(\alpha X)) + i\mathbb{E}(\sin(\alpha X)),$$

where the integrals are always finite since $|\cos|$ and $|\sin|$ are bounded by 1.

Example 12.3. *Generating functions of the normal distribution.* Recall that a normal-distributed random variable $Z \sim N(\mu, \sigma^2)$ for parameters $\mu \in \mathbb{R}, \sigma^2 > 0$ has density

$$f(z) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}, \quad z \in \mathbb{R}.$$

We would like to find the moment generating function of Z. But first we consider $X \sim N(0,1)$, then

$$\begin{aligned}
M_X(\alpha) = \mathbb{E}(e^{\alpha X}) &= \int_{-\infty}^{\infty} e^{\alpha x} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2 - 2\alpha x + \alpha^2 - \alpha^2)}\, dx \\
&= e^{\frac{1}{2}\alpha^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-\alpha)^2}\, dx \\
&= e^{\frac{1}{2}\alpha^2},
\end{aligned}$$

since we recognise the second factor as the integral of the density of $N(0, \alpha)$-distributed random variable, which is therefore equal to 1.

To calculate the characteristic function of X, we would like to do the following calculation

$$\varphi_Z(\alpha) = \mathbb{E}(e^{i\alpha X}) = M_X(i\alpha) = e^{\frac{1}{2}(i\alpha)^2} = e^{-\frac{1}{2}\alpha^2}.$$

In general, this is not allowed. However, in this special case it gives the right answer (the keyword for a proof is: analytic continuation).

If $Z \sim N(\mu, \sigma^2)$ is a general normal random variable, then $Z = \mu + \sigma X$ for a standard-normal random variable $X \sim N(0, 1)$. In particular, we have

$$M_Z(\alpha) = \mathbb{E}(e^{\alpha(\mu+\sigma X)}) = e^{\alpha\mu}\mathbb{E}(e^{\alpha\sigma X}) = e^{\alpha\mu}M_X(\alpha\sigma) = e^{\alpha\mu+\frac{1}{2}\alpha^2\sigma^2}.$$

Moreover, for the characteristic function, we get

$$\varphi_Z(\alpha) = \mathbb{E}(e^{i\alpha(\mu+\sigma X)}) = e^{i\alpha\mu}\mathbb{E}(e^{i\alpha\sigma X}) = e^{i\alpha\mu}\varphi_X(\alpha\sigma) = e^{i\alpha\mu-\frac{1}{2}\alpha^2\sigma^2}.$$

Similarly as for PGFs, we can show the following for MGFs:

Lemma 12.4 (Properties of MGFs). *(i) If $M(\alpha)$ is defined for α from an open interval containing 0, then for $k \in \mathbb{N}$*

$$\mathbb{E}(X^k) = M_X^{(k)}(0),$$

where $M_X^{(k)}$ denotes the kth derivative of M_X.

(ii) If X, Y are independent, then for all α such that the right hand side is defined

$$M_{X+Y}(\alpha) = M_X(\alpha)M_Y(\alpha).$$

Instead of continuing to explore the properties of M, we rather consider characteristic functions, where we don't have to worry about existence.

Proposition 12.5 (Properties of CFs). *(a) A characteristic function φ uniquely determines the distribution function $F_X(x) := \mathbb{P}(X \leq x)$.*

(b) If X and Y are independent, then

$$\varphi_{X+Y}(\alpha) = \varphi_X(\alpha)\varphi_Y(\alpha) \quad \text{for all } \alpha \in \mathbb{R}.$$

(c) If for some $k \in \mathbb{N}$, we have that $\mathbb{E}\big(|X|^k\big) < \infty$, then

$$\varphi_X(t) = \sum_{j=0}^{k} \frac{\mathbb{E}(X^j)}{j!}(it)^j + o(t^k)$$

as $t \to 0$ and

$$\varphi^{(k)}(0) := \frac{\mathbb{P}artial^k}{\mathbb{P}artial\alpha^k}\varphi_X(\alpha)\bigg|_{\alpha=0} = i^k\mathbb{E}(X^k)$$

Recall that we write $g(t) = o(t^k)$ as $t \to 0$ if $\lim_{t\to 0, t\neq 0} \frac{g(t)}{t^k} = 0$.

Proof. Part (c) is essentially Taylor's theorem for a function of a complex variable. $\square$

Theorem 12.6 (Continuity theorem). *Let $F, F_1, F_2, \dots$ be distribution functions with CFs $\varphi, \varphi_1, \varphi_2, \dots$*

(i) If $F_n(x) \to F(x)$ for all points x at which F is continuous, then $\varphi_n(\alpha) \to \varphi(\alpha)$ $\forall \alpha \in \mathbb{R}$;

(ii) Conversely, if $\varphi(\alpha) = \lim_{n\to\infty} \varphi_n(\alpha)$ exists for all $\alpha \in \mathbb{R}$ and φ is continuous at $\alpha = 0$, then φ is the CF of some distribution function F and $F_n(x) \to F(x)$ for all x where F is continuous.

For a proof, see [GS01, Theorem 5.9.5].

Recall that if $X_1, X_2, \dots$ are i.i.d. random variables with $\mathbb{E}(|X_i|) < \infty$, then

$$\frac{1}{n}\sum_{i=1}^n X_i \to \mathbb{E}(X_1) \quad \text{almost surely.}$$

Rewriting we have that $\sum_{i=1}^n X_i \approx n\mathbb{E}(X_1)$. The next theorem, the *central limit theorem* tells us that if $\mathbb{E}(X_i^2) < \infty$, then the next term in this expansion is of order $\sqrt{n}$. Moreover, the prefactor is random and normally distributed, so it does *not* depend on the distribution of the X_i that we are started with.

Theorem 12.7 (Central limit theorem). *Let $X_1, X_2, \dots$ be independent and identically distributed random variables with $\mathbb{E}(X_i) =: \mu$, $Var(X_i) =: \sigma^2 < \infty$, then with*

$$S_n := X_1 + X_2 + \cdots + X_n = \sum_{i=1}^n X_i$$

we have that

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \xrightarrow{\mathcal{D}} Z$$

where $Z \sim N(0,1)$. That is, for each $x \in \mathbb{R}$,

$$\mathbb{P}\left(\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \leq x\right) \longrightarrow \Phi(x) := \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\,\mathrm{d}y$$

Note the normalization of the sum is chosen so that the expectation of the left hand side is

$$\mathbb{E}\left(\frac{1}{\sqrt{n\sigma^2}}\left(\sum_{i=1}^n X_i - n\mathbb{E}(X_1)\right)\right) = 0,$$

and further

$$\mathrm{Var}\left(\frac{1}{\sqrt{n\sigma^2}}\left(\sum_{i=1}^n X_i - n\mathbb{E}(X_1)\right)\right) = \frac{1}{n\sigma^2}\mathrm{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n\sigma^2}\sum_{i=1}^n \mathrm{Var}(X_i) = 1,$$

where we used the independence to interchange sum and taking variance.

Proof. W.l.o.g we may assume that $\mu = 0$ and $\sigma^2 = 1$ (otherwise consider $Y_i = \frac{X_i - \mu}{\sqrt{\sigma^2}}$). We want to use the continuity theorem, Theorem 12.6, and so it will suffice to show that the characteristic function

$$\varphi_n(\alpha) := \mathbb{E}(e^{i\alpha \frac{S_n}{\sqrt{n}}}),$$

satisfies $\varphi_n(\alpha) \to \varphi_Z(\alpha)$ for all α, where

$$\varphi_Z(\alpha) := \mathbb{E}\big(e^{i\alpha Z}\big) = e^{-\alpha^2/2}$$

is the characteristic function of $Z \sim N(0,1)$.
Note that $\varphi_1(\alpha) = \mathbb{E}\big(e^{i\alpha X_1}\big)$. Then, by the assumption of independence we have

$$\varphi_n(\alpha) = \mathbb{E}\left(e^{i\alpha S_n/\sqrt{n}}\right) = \mathbb{E}\left(e^{i\alpha(X_1+\cdots+X_n)/\sqrt{n}}\right) = \varphi_1\left(\frac{\alpha}{\sqrt{n}}\right)^n$$

Now, by Proposition 12.5 we have

$$\begin{aligned}\varphi_n(\alpha) &= \left\{1 + \frac{i\alpha}{\sqrt{n}}\mathbb{E}(X_1) + \frac{(i\alpha)^2}{2n}\mathbb{E}(X_1^2) + o\left(\frac{\alpha^2}{n}\right)\right\}^n \\ &= \left\{1 - \frac{\alpha^2}{2n} + o\left(\frac{\alpha^2}{n}\right)\right\}^n \\ &\to e^{-\alpha^2/2} = \varphi_Z(\alpha) \text{ as } n \to \infty,\end{aligned}$$

where we used that $(1 + \frac{x}{n}(1 + o(1)))^n \to e^x$ for all x. □

The following material in this chapter is not examinable!

How accurate is the central limit theorem?

Theorem 12.8 (Berry-Esseen)**.** *If $X_1, X_2, \ldots$ are i.i.d. with $\mathbb{E}(X_i) = 0$, $\mathrm{var}(X_i) = \sigma^2$ and $\mathbb{E}|X_1|^3 < \infty$, then for all x,*

$$\left|\mathbb{P}\left(\frac{S_n}{\sqrt{n\sigma^2}} \le x\right) - \Phi(x)\right| \le \frac{3\mathbb{E}|X_1|^3}{\sigma^3\sqrt{n}}$$

Note that the rate $1/\sqrt{n}$ cannot be improved in general, but the constant 3 can be slightly optimised. This result gives the maximum error in the normal approximation of the CLT and it works for finite n and is not a purely asymptotic statement.

Example 12.9. Toss a fair coin 100 times. What is the probability of 65 or more heads?

Solution. We have that $n = 100$, $\mu = \frac{1}{2}$, $\sigma^2 = \frac{1}{2}\frac{1}{2} = \frac{1}{4}$. So $X_i \sim B(1, 1/2)$ are i.i.d. and $S_n \sim B(n, 1/2)$. By Chebychev's inequality,

$$\begin{aligned}\mathbb{P}\big(|S_{100} - 50| > 15\big) &\le (15)^{-2}\mathrm{Var}(S_{100}) \\ &= \frac{1}{225} \cdot 100 \cdot \frac{1}{4} = \frac{1}{9}\end{aligned}$$

Hence $\mathbb{P}(\text{at least 65 heads}) \leq 1/18$ by symmetry. On the other hand, using the CLT,

$$\frac{S_{100} - 50}{\sqrt{25}} \text{ is approximately} N(0,1)\text{-distributed.}$$

So

$$\mathbb{P}(S_{100} - 50 \geq 15) = \mathbb{P}\left(\frac{S_{100} - 50}{\sqrt{25}} \geq 3\right) \approx \mathbb{P}(Z \geq 3) = 0.0014$$

where $Z \sim N(0,1)$. We can see the accuracy of this by the Berry-Essen theorem; this says that $\forall x \in \mathbb{R}$,

$$\left|\mathbb{P}\left(\frac{\sum_{i=1}^{n}(X_i - 1/2)}{\sqrt{100(1/2)^2}} \leq x\right) - \Phi(x)\right| \leq \frac{3\mathbb{E}|X_1 - 1/2|^3}{(1/8)\sqrt{100}} = \frac{3}{10}$$

where $\mathbb{E}|X_1 - 1/2|^3 = 1/8$ is implied by $\mathbb{P}(X_1 = 1) = 1/2$ and $\mathbb{P}(X_1 = 0) = 1/2$. So,

$$\left|\mathbb{P}(S_{100} \geq 65) - 0.0014\right| \leq 0.3$$

i.e. the worst approximation is $\mathbb{P}(S_{100} \geq 65) \leq 0.3014$.

References

[GS01] G. Grimmett and D. Stirzaker. *Probability and random processes.* Oxford University Press, third edition, 2001.

A Important distributions

Discrete distributions

Name	Probability mass function	$\mathbb{E}(X)$	$\mathrm{Var}(X)$	PGF
Bernoulli $\mathrm{Ber}(p)$	$\mathbb{P}(X=1) = p = 1 - \mathbb{P}(X=0)$	p	$p(1-p)$	$p\theta + (1-p)$
Binomial $\mathrm{B}(n,p)$	$\binom{n}{k} p^k (1-p)^{n-k}$, $k \in \{0, \dots, n\}$	np	$np(1-p)$	$(p\theta + (1-p))^n$
Poisson $\mathrm{Po}(\lambda)$	$e^{-\lambda} \frac{\lambda^k}{k!}, k = 0, 1, 2, \dots$	λ	λ	$\exp\{\lambda(\theta - 1)\}$

Continuous distributions

Name	Probability density function	Expectation	Variance[3]
Exponential $\mathrm{Exp}(\lambda)$	$\lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Uniform $\mathrm{Unif}[a,b]$	$\frac{1}{b-a} \mathbb{1}_{[a,b]}(x), x \in \mathbb{R}$	$\frac{a+b}{2}$	$\frac{1}{12}(b-a)^2$
Normal $\mathrm{N}(\mu, \sigma^2)$	$\dfrac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\dfrac{(x-\mu)^2}{2\sigma^2} \right\}, x \in \mathbb{R}$	μ	σ^2

[3] Not examinable